PRODUCT & TECHNICAL SPECIFICATION

# Voice Agent Self-Service Portal

A self-service SaaS where a small or medium service business builds, tests and publishes an AI phone agent in under 15 minutes, then runs it on prepaid credits bought from us. Retell AI is the underlying platform and is never exposed to the customer.

| | |
|---|---|
| WORKING NAME | TBD (placeholder: “the Portal”) |
| OWNER | Pramod (CTO) |
| DATE | 10 Sep 2026 |
| VERSION | v0.2 — expanded: authentication, screen-by-screen UI, prompt system, dashboard, FE/BE breakdown |
| STATUS | Draft for review → .docx after sign-off |
| PLATFORM | Retell AI (REST API + webhooks) — never exposed to the customer |

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HOW TO READ THE SCREEN SPECS (§6, §8, §10)

Every screen is described with the same blocks so a developer can build it without asking:

**Route / step / access —** URL, position in the flow, which roles can see it.

**Purpose —** one sentence.

**Layout —** wireframe of the key screens.

**Fields —** every input: type, required, validation, default, where it is stored.

**Actions —** every button and link: what the frontend does, which API it calls, what the backend does (including Retell and Stripe calls), success and failure behaviour.

**States —** loading, empty, error, disabled conditions.

# 01 Summary

A self-service SaaS where a small/medium service business builds, tests and publishes an AI phone agent in under 15 minutes, then runs it on **prepaid credits bought from us.** We wrap Retell AI (voice infra, LLM, TTS, telephony) with:

1. A guided **"brief → agent"** wizard — the customer describes their business and goal in plain language; we generate and manage the prompt.
2. A **test-and-tune loop —** browser test call, live transcript, plain-language feedback, re-test, publish.
3. **Phone provisioning —** buy a number through us, forward an existing number, or (advanced) connect via SIP.
4. A **prepaid wallet —** credit packs via Stripe, per-second metering from Retell webhooks, hard stop at zero.

FOUR CORE PRODUCT BETS

**Prepaid, not postpaid.** Customer buys a pack; agent goes offline at zero balance. No surprise invoices, no collections risk.

**Curated stack.** We choose LLM + voice tiers (Standard / Premium). Customer never sees "GPT-4.1 vs Claude". Predictable COGS.

**Brief, not prompt.** The raw prompt is hidden behind a structured "Agent Brief". Advanced users can open it.

**Test before you pay.** Free browser-only test minutes; phone goes live only after first purchase.

# 02 Goals and non-goals

## Goals (MVP)

- Sign-up → published inbound agent on a real phone number in < 15 minutes, no human involvement.
- Prepaid credit purchase, balance, alerts, auto top-up.

- Accurate per-second metering reconciled against Retell's own call cost.
- Clear, boring pricing: one unit (credit), two agent tiers, published pack prices.
- Multi-language support (primary + additional languages) from day one — matters for Quebec/Canada (fr-CA) and NA Spanish.
- Every screen usable by a non-technical owner on a laptop or phone; no field without a default or an example.

### Non-goals (MVP)

- Outbound campaigns / auto-dialler (Phase 2 — compliance-gated).
- Custom tool/function calling, live calendar/CRM integrations (Phase 2).
- Number porting (Phase 3).
- White-label / agency reseller tier (Phase 3).
- Chat/SMS agents (Phase 3 — Retell supports it; keep the data model channel-aware).

# 03 Target customer and use cases

**ICP:** owner-operated service businesses in North America (clinics, dental, home services, salons, legal intake, real estate, auto shops, restaurants) that miss calls and don't have a developer. Same ICP as the Metacognixion "capacity architecture" thesis — this is the cheapest on-ramp into that portfolio.

LAUNCH TEMPLATES (FULL DETAIL IN §9)

| TEMPLATE | DIRECTION | WHAT IT DOES |
|---|---|---|
| AI Receptionist | Inbound | Answers, takes messages, answers FAQs, transfers if needed |
| Appointment Request | Inbound | Collects preferred date/time + details; confirms a human will call back (no live calendar in MVP) |
| After-hours Answering | Inbound (forwarded/overflow) | Handles calls when closed; emergency escalation rules |
| Lead Qualification | Inbound | Asks qualifying questions, scores lead, captures contact |
| FAQ / Support | Inbound | Answers from a knowledge base (Growth+ plans) |
| Restaurant Reservations | Inbound | Takes reservation requests, hours/menu FAQs, large-party handoff |
| Custom | Inbound | Blank brief with guided fields |

# 04 Retell AI cost basis

Verified 10 Sep 2026

Source: retellai.com/pricing (pay-as-you-go, no platform fee, $10 free credits, 20 concurrent calls included). All per-minute unless noted. Retell bills **per second**, only for connected calls; AI charges stop after a transfer (telephony continues).

## 4.1 Per-minute components

| COMPONENT | OPTION | COST / MIN |
|---|---|---|
| Voice infrastructure (mandatory) | Retell | **$0.055** |
| Text-to-speech | Retell platform / Minimax / Fish / Cartesia / OpenAI | $0.015 |
| | ElevenLabs | $0.040 |

| COMPONENT | OPTION | COST / MIN |
|---|---|---|
| LLM (standard tier) | GPT-5 nano | $0.003 |
| | GPT-4.1 nano | $0.004 |
| | Gemini 2.5 Flash Lite | $0.006 |
| | GPT-5.4 nano | $0.010 |
| | GPT-5 mini | $0.012 |
| | GPT-4.1 mini | $0.016 |
| | Claude 4.5 Haiku | $0.025 |
| | GPT-5.4 mini | $0.036 |
| | GPT-5 / GPT-5.1 | $0.040 |
| | GPT-4.1 | $0.045 |
| | GPT-5.2 | $0.056 |
| | Claude 4.5 / 4.6 / 5 Sonnet | $0.080 |
| | GPT-5.4 | $0.080 |
| | GPT-5.5 | $0.160 |
| LLM “fast tier” (lower latency) | ~2x standard tier (e.g. GPT-4.1 fast = $0.0675) | |
| Speech-to-speech LLM | GPT Realtime mini / GPT Realtime | $0.07 / $0.345 |
| Telephony | Retell-purchased Twilio/Telnyx number, US/CA | $0.015 |
| | Customer’s own SIP trunk (BYO) | $0.00 (customer pays carrier) |
| | Web call (browser widget) | $0.00 |
| Knowledge base usage | | +$0.005 |
| Advanced denoising | | +$0.005 |

| COMPONENT | OPTION | COST / MIN |
|---|---|---|
| Safety guardrails | | +$0.005 |
| PII removal | | +$0.010 |
| AI QA | first 100 min free | +$0.10 (do **not** enable by default) |
| Batch calling | | +$0.005 / dial |
| Branded caller ID | | +$0.10 / outbound call |
| SMS | | $0.01 / msg |

## 4.2 Monthly / fixed items

| ITEM | COST | NOTE |
|---|---|---|
| Retell phone number | $2.00 / number / month | $0 for imported/custom numbers |
| Concurrency | first 20 free, then **$8 / concurrent call / month** | **Shared across our whole Retell account = all tenants** |
| Knowledge base | first 10 free, then **$8 / KB / month** | **Shared across account** = effectively $8/tenant/month once we pass 10 |
| Verified phone number (anti-spam) | $10 one-time | outbound only |
| Retell SMS | $20 / month | Phase 3 |

### 4.3 Our curated stacks (COGS)

| | STANDARD TIER | PREMIUM TIER |
|---|---|---|
| Voice infra | $0.055 | $0.055 |
| TTS | Retell platform / Cartesia — $0.015 | ElevenLabs — $0.040 |
| LLM | GPT-5 mini — $0.012 (fallback GPT-4.1 mini $0.016) | GPT-4.1 — $0.045 (or Claude 4.5 Haiku $0.025 if quality holds) |
| Telephony (Retell number) | $0.015 | $0.015 |
| KB (when used) | $0.005 | $0.005 |
| **COGS / min (phone, Retell number)** | **≈ $0.10–0.11** | **≈ $0.16** |
| COGS / min (BYO SIP or web widget) | ≈ $0.09 | ≈ $0.145 |

ACTION

*Validate model quality and latency on bench.retellai.com before locking Standard = GPT-5 mini. If it fails the receptionist template, step up to GPT-4.1 mini (+$0.004) — margin barely moves.*

# 05 Business model — prepaid credits

### 5.1 Unit

- **1 credit = 1 minute of a Standard-tier agent on the phone.**
- **Premium tier = 1.5 credits / minute.**
- Web-widget calls (no telephony): same rate in MVP (simplicity); revisit later.
- Metered **per second**, pro-rated (e.g. 2 min 30 s Standard = 2.50 credits). Balance shown to 2 decimals.
- Only connected calls are charged (mirrors Retell). Voicemail: charged while agent is on the line (mirrors Retell).
- Transferred calls: **MVP bills full agent rate for the whole call** (simple). Phase 2: bill transferred leg at 0.25 credit/min (telephony-only) using `transfer_bridged` / `transfer_ended` events.

## 5.2 Packs (USD, one-off, prepaid)

| PACK | PRICE | CREDITS | $/CREDIT | STD MARGIN (COGS $0.11) | PREMIUM MARGIN (1.5 CR/MIN VS $0.16) | CONCURRENCY CAP | KB INCL. |
|---|---|---|---|---|---|---|---|
| Starter | $49 | 150 | $0.327 | 66% | 67% | 2 | No (FAQ text goes in brief) |
| Growth | $149 | 500 | $0.298 | 63% | 64% | 3 | 1 |
| Scale | $399 | 1,500 | $0.266 | 59% | 60% | 5 | 3 |
| Pro | $999 | 4,500 | $0.222 | 50% | 52% | 10 | 5 |
| Volume | custom | 10,000+ | ~$0.18 | 39% | 41% | negotiated | negotiated |

Market check: competitors quote roughly $0.25–0.50/min all-in for SMB voice agents; Retell-based agencies commonly resell at $0.30–0.40. We sit at the affordable end with a clear floor.

The **plan tier** (Starter/Growth/Scale/Pro) is set by the largest pack bought in the last 12 months; it governs concurrency cap, KB allowance and feature gates. Credits from any pack pool into one balance.

## 5.3 Recurring deductions from the wallet

- **Phone number rental (Retell-purchased):** 15 credits / number / month, deducted on the purchase-date anniversary. Forwarded existing numbers still need one purchased number (the forward target). BYO SIP numbers: 0 credits.
- **Knowledge base:** included per plan above; extra KB = 30 credits / month (covers Retell's $8 once we exceed the 10 free).
- **Concurrency above plan cap:** not purchasable in MVP — customer upgrades pack. Platform buys Retell slots ($8/mo) when platform-wide peak > 16 (80% of 20).

## 5.4 Wallet rules

- Credits expire **12 months** after purchase (FIFO consumption). Non-refundable, non-transferable.
- **Low-balance alerts** at 20% and 5% of last pack size, and at 0 (email + in-app banner + optional SMS to owner).
- **Auto top-up** (opt-in): when balance < X credits, charge saved card for chosen pack. Default suggested: on, at 10% of last pack.
- **Grace overdraft:** up to 10 credits so an in-progress call is never cut mid-sentence; next top-up settles it first. Hard stop after that (see §14).
- **Free trial:** 20 credits, **browser test calls only** (no phone number until first purchase). Max exposure ≈ $1.80/sign-up. No card required for trial.

- Invoices generated per purchase (Stripe), with company details editable in settings.

### 5.5 Decisions needed on pricing

1. Credit unit: **credits with 1.5x premium multiplier** (recommended) vs. plain "minutes" with premium gated to Scale+ plans.
2. Pack price points above are proposals — confirm CAD vs USD billing, and whether Metacognixion (Canada) or Gritbyte (India) is the merchant of record for Stripe.
3. Whether to expose a "Fast tier" (lower latency, ~2x LLM cost) as a Premium+ option. Recommendation: no for MVP.

# 06 Authentication, accounts and workspaces

### 6.1 Model

- **User** = a person (email). **Workspace** (= tenant) = a business. A user can belong to many workspaces; a workspace has many users via **memberships** with a role.
- Everything billable (credits, agents, numbers, calls) belongs to the workspace, never the user.

| CAPABILITY | OWNER | ADMIN | EDITOR | VIEWER |
|---|---|---|---|---|
| Buy credits, auto top-up, payment methods, invoices | ✓ | ✓ | – | – |
| Invite/remove members, change roles | ✓ | ✓ (not Owner) | – | – |
| Transfer ownership, delete workspace | ✓ | – | – | – |
| Create/edit/publish agents, run test calls | ✓ | ✓ | ✓ | – |
| Buy/configure/release numbers | ✓ | ✓ | – | – |
| View calls, transcripts, recordings, analytics | ✓ | ✓ | ✓ | ✓ |
| Change workspace settings (name, hours, retention) | ✓ | ✓ | – | – |
| Security settings (MFA for self, sessions) | ✓ | ✓ | ✓ | ✓ |

Exactly one Owner per workspace. The sign-up user becomes Owner.

### 6.2 Provider and mechanics

- **Managed auth provider** (recommendation: Clerk for Next.js speed; Auth0 or Supabase Auth acceptable). We do not hand-roll password storage, email verification, OAuth or MFA.

- Methods: **email + password**, **magic link** (email), **Google**, **Microsoft** (Outlook/M365 users are common in NA SMBs). Apple sign-in later for the mobile web.
- Email verification mandatory before any test call or purchase.
- **MFA:** TOTP (authenticator app) optional for all; **required for Owner/Admin once auto top-up is enabled** or balance > 1,000 credits (money at risk). Recovery codes shown once.
- **Sessions:** provider-issued session cookie, HttpOnly, Secure, SameSite=Lax, 30-day sliding expiry, revocable per device from Settings ▸ Security. Short-lived JWT (≤ 15 min) is exchanged by the frontend for backend calls; backend verifies signature (JWKS) and reads `sub` (user id).
- **Tenant context:** every backend request carries `X-Workspace-Id`. Middleware: verify JWT → load user → load membership for (user, workspace) → deny if missing → attach `ctx = {userId, workspaceId, role}`. All queries are scoped by `workspace_id` (Postgres RLS as a second guard).
- **CSRF:** SameSite cookie + origin check on state-changing requests. **Rate limiting:** login 5 failures / 15 min per email+IP, then CAPTCHA; sign-up 3 / hour / IP; magic links single-use, 15-min TTL.
- **Disposable email domains blocked** at sign-up (open-source list). Free-mail domains allowed (many SMBs use Gmail).
- **Audit log** (workspace-scoped): login, member changes, role changes, publish, number bought/released, purchases, auto top-up changes, data deletion. Visible to Owner/Admin in Settings.

## 6.3 Auth screens

### A1 — Sign up

Route `/signup` · Public.

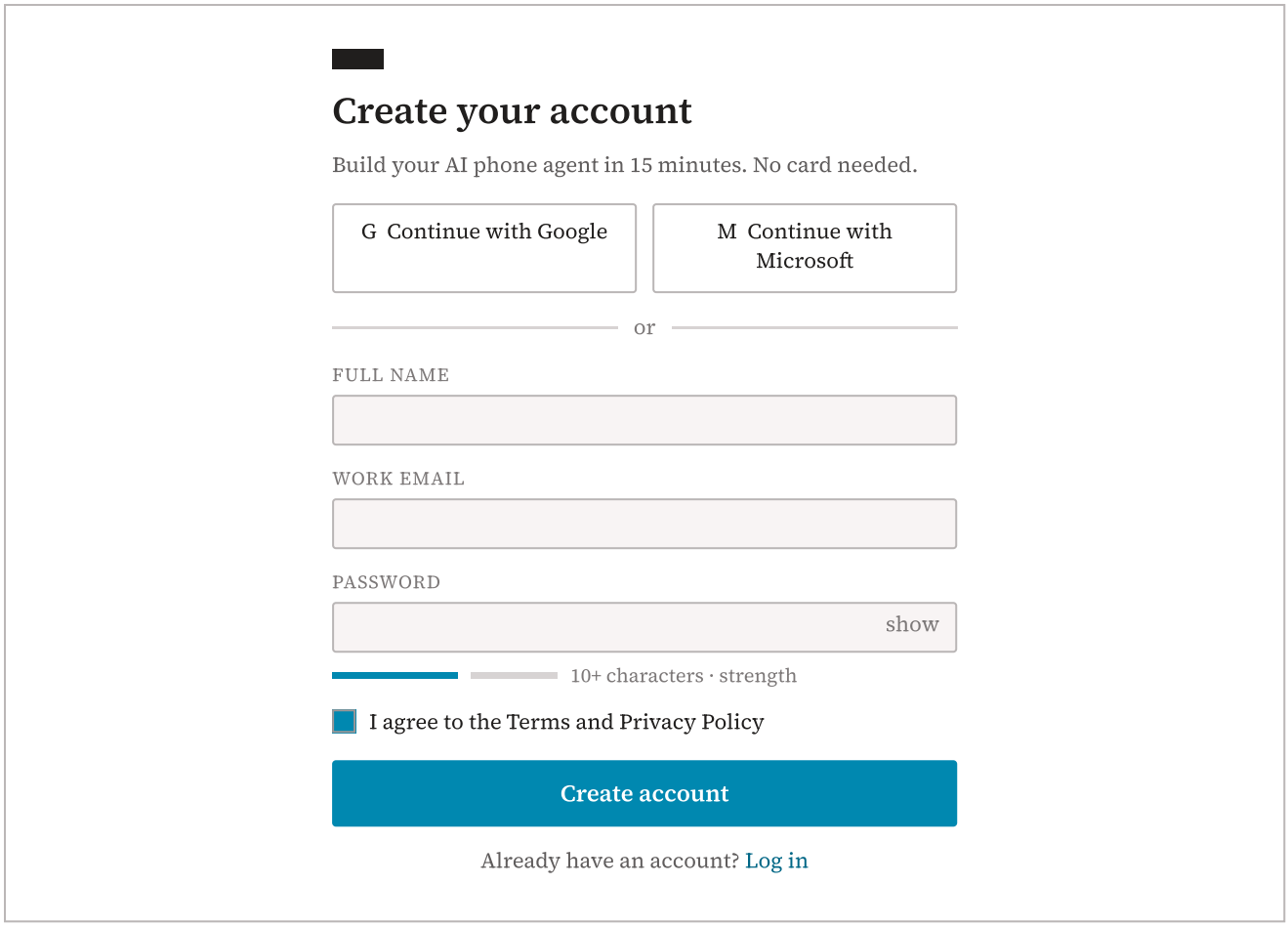


Figure 1 — A1 Sign up. Single-column card: federated identity first, then the email form; one primary action.

| FIELD | TYPE | REQ | VALIDATION | NOTES |
|---|---|---|---|---|
| Full name | text | ✓ | 2–80 chars | stored on `users.name` |
| Work email | email | ✓ | RFC format; not disposable; unique | lower-cased |
| Password | password | ✓ (unless SSO) | ≥ 10 chars, not in breached list (provider feature) | strength meter |
| Terms checkbox | bool | ✓ | must be checked | timestamp stored (`users.tos_accepted_at`) |

| ACTION | FRONTEND | API / BACKEND |
|---|---|---|
| Continue with Google/Microsoft | OAuth redirect via provider | Provider creates user; webhook `user.created` → backend creates `users` row; redirect `/verify-email` (skipped if provider reports verified) → `/setup/workspace` |
| Create account | Validate with zod; disable button; show inline errors | Provider sign-up → `user.created` webhook → `users` row → send verification code → route `/verify-email` |
| Log in link | → `/login` | — |

**States:** field-level errors inline; “email already registered” → offer login/magic link; network error → retry toast.

### A2 — Verify email

Route `/verify-email` · Authenticated (unverified).

6-digit code input (auto-advance, paste-aware) + “Resend code” (60-s cooldown, max 5/hour) + “Use a different email”. On success → `/setup/workspace` (if no workspace) else `/app`.

### A3 — Workspace setup

Route `/setup/workspace` · Authenticated, no workspace yet.

| FIELD | TYPE | REQ | VALIDATION | DEFAULT |
|---|---|---|---|---|
| Business name | text | ✓ | 2–80 chars | — |
| Country | select (US, CA at launch; others “coming soon”) | ✓ | — | from IP geolocation |
| Time zone | select (IANA) | ✓ | — | browser TZ |
| Business phone (for alerts) | tel | – | E.164 | — |
| Website | url | – | — | used later to pre-fill the brief (Phase 2 crawl) |

Button **Continue** → `POST /workspaces` → creates `tenants` row, membership (Owner), grants 20 trial credits (`credit_ledger` reason=`trial`), default notification settings → redirect `/onboarding/template`.

### A4 — Log in

Route `/login` · Public. Email + password, “Forgot password?”, “Email me a magic link”, Google, Microsoft. After login: if MFA enrolled → A5; if 1 workspace → `/app`; if many → A7; if none → A3. “Remember this device” 30 days.

### A5 — MFA challenge

6-digit TOTP or recovery code. 5 attempts then 15-min lock.

### A6 — Forgot / reset password

Email → provider sends reset link (15-min TTL) → new password form → all other sessions revoked → login.

### A7 — Workspace switcher

Route `/select-workspace`. List of workspaces (name, role, balance pill). Also available from top bar. Selecting sets `X-Workspace-Id` in the frontend store and cookie `ws`.

### A8 — Accept invite

Route `/invite/[token]`. Shows "You've been invited to {workspace} as {role}". If not logged in → sign up/login with email pre-filled and locked to invitee address. Button **Join workspace** → `POST /invites/:token/accept` → membership created → `/app`. Token single-use, 7-day TTL; wrong-email login shows explicit error.

## 6.4 Onboarding gating (route guards)

| CONDITION | REDIRECT |
|---|---|
| Not authenticated | `/login?next=…` |
| Email unverified | `/verify-email` |
| No workspace | `/setup/workspace` |
| Workspace has no published agent and onboarding not dismissed | `/onboarding/*` (resume at last completed step, stored in `tenants.onboarding_step`) |
| Viewer tries to open a write route | `/app` with toast "You don't have permission" |

# 07 Information architecture, navigation and design system

## 7.1 Sitemap

PUBLIC & ACCOUNT

`/signup` `/login` `/verify-email` `/forgot-password` `/reset-password` `/invite/[token]`
`/select-workspace` `/setup/workspace`

/ONBOARDING — THE WIZARD

| | | |
|---|---|---|
| `/template` | W1 | Choose what your agent does |
| `/brief` | W2 | Tell us about your business → Generate |
| `/voice` | W3 | Voice & languages |
| `/knowledge` | W4 | Knowledge (optional) |
| `/test` | W5 | Test & improve |
| `/phone` | W6 | Phone number |
| `/activate` | W7 | Buy credits & go live → /onboarding/live (success) |

/APP — THE DASHBOARD

| | | |
|---|---|---|
| `/app` | D1 | Home |
| `/app/calls` | D2 | Calls (list) · /app/calls/[id] (detail drawer/page) |
| `/app/agents` | D3 | Agents (list) · /app/agents/[id]/(overview\|brief\|voice\|knowledge\|test\|versions\|settings) |
| `/app/numbers` | D4 | Numbers (list) · /app/numbers/[id] |
| `/app/wallet` | D5 | Wallet & billing (balance\|packs\|auto-topup\|ledger\|invoices\|payment-methods) |
| `/app/settings` | D6 | Workspace · Team · Notifications · Security · Data & privacy · Audit log |
| `/app/help` | D7 | Help |
| `/admin/*` | D8 | Internal ops console (staff only, separate auth role) |

Figure 2 — Sitemap. Three route groups: account, the seven-step wizard, the dashboard.

## 7.2 Global shell (all `/app` routes)

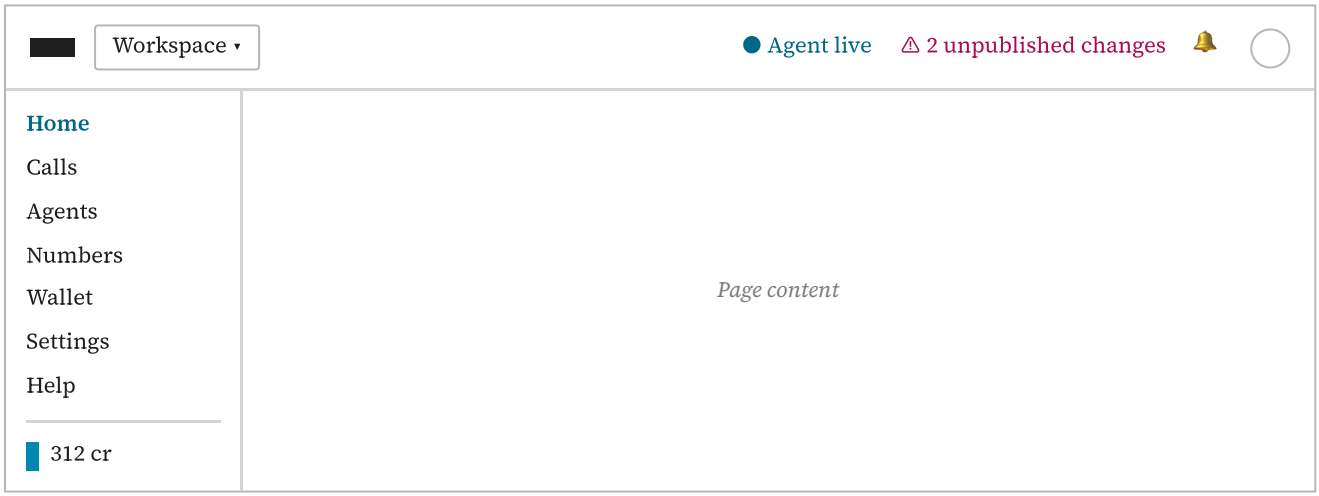


Figure 3 — Global shell. Left nav, workspace switcher and status chips in the top bar; the balance pill is pinned to the foot of the nav (credits, ≈ minutes, “Top up”).

- **Balance pill** turns amber < 20% of last pack, red at ≤ 0 (“Agent paused — top up”).

- **Unpublished changes** chip appears when any agent's draft version ≠ published version; click → agent's Test tab with a **Publish** button.
- **Notifications bell:** low balance, new messages captured, number provisioning done, failed auto top-up, member joined.
- Mobile: left nav collapses to a bottom tab bar (Home, Calls, Agents, Wallet, More).

### 7.3 Design system

- Tailwind + shadcn/ui components; 8-px spacing grid; one primary colour + semantic (success/warn/danger); Inter font.
- Every form field: label, helper text or example placeholder, inline validation on blur, error text below the field. Never a bare input.
- Primary action bottom-right on desktop, sticky bottom on mobile. One primary button per screen.
- Wizard shell: stepper at top (7 steps, completed = ✓, current = highlighted, future = grey but clickable if previously completed), "Save & exit" top-right (draft persists), **Back** and **Continue** bottom.
- Tone of copy: plain English, no jargon ("voice" not "TTS", "what your agent knows" not "knowledge base"). fr-CA translation from launch (next-intl).
- Accessibility: WCAG 2.1 AA, full keyboard navigation, live-region announcements for call state.

# 08 Onboarding wizard — screen-by-screen

Wizard state is **server-persisted**: each step writes to the draft agent (`agents` + `agent_versions`) so a refresh, another device, or "Save & exit" never loses work. `tenants.onboarding_step` tracks the resume point.

### W1 — Choose what your agent does

Route `/onboarding/template` · Step 1 of 7 · Owner/Admin/Editor. **Purpose:** pick a template so every later screen is pre-filled.

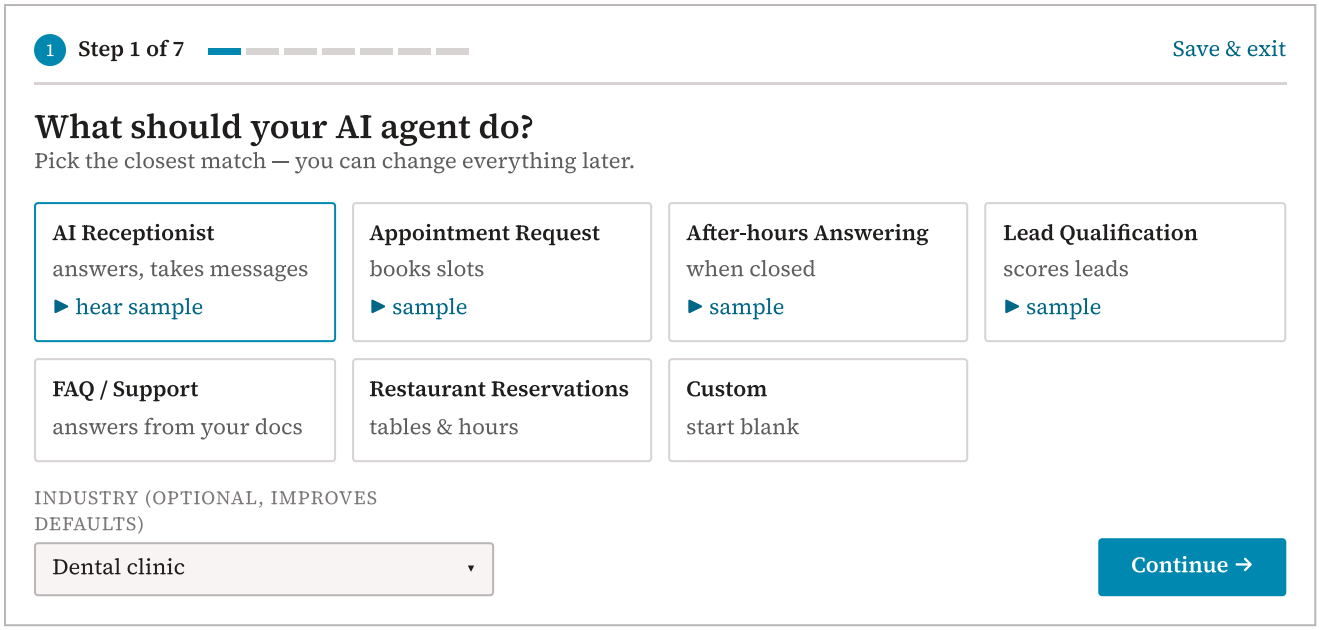


Figure 4 — W1 Template picker. Seven radio cards, industry select, one primary action bottom-right.

| ELEMENT | TYPE | REQ | NOTES |
|---|---|---|---|
| Template card | single-select radio cards | ✓ | selected card highlighted; "▶ hear sample" plays a 20-s pre-recorded call (static asset, no credits) |
| Industry | searchable select (≈ 40 options) | – | drives brief examples, default voice and consent rules |

| ACTION | FRONTEND | BACKEND |
|---|---|---|
| Continue | requires a selection | `POST /agents {template_id, industry}` → creates `agents` (status `draft`, tier from template default) + `agent_versions` `v1` with `brief_json` pre-filled from template → returns `agent_id`; sets `onboarding_step=2` |
| Save & exit | — | draft already saved; go to `/app` with checklist |

**States:** templates load from a static config (no network); Continue disabled until a card is selected.

## W2 — Tell us about your business (the Brief)

Route `/onboarding/brief` · Step 2 of 7 · Owner/Admin/Editor. **Purpose:** capture everything needed to generate the agent in plain language.

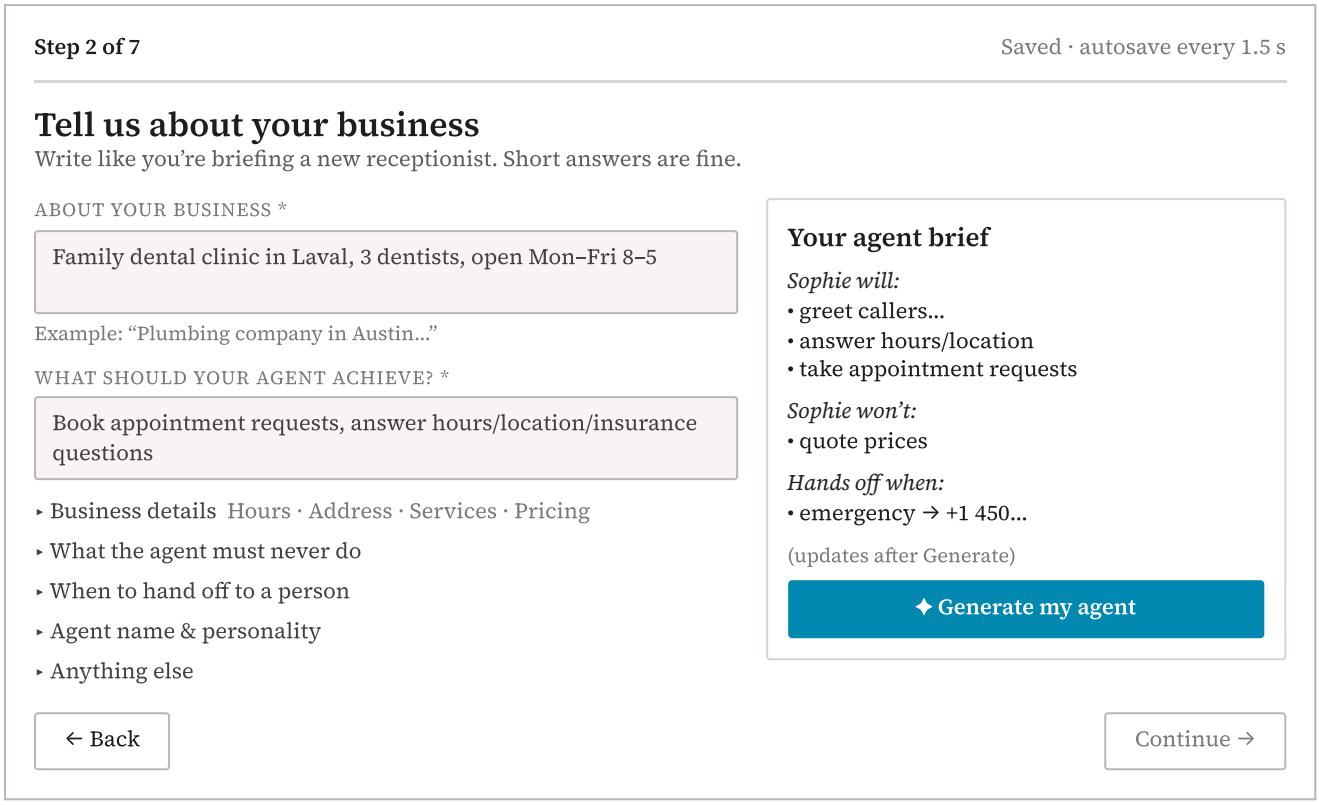


Figure 5 — W2 Brief. Scrolling form left, sticky brief preview right; Continue stays disabled until a successful Generate.

FIELDS (ALL STORED IN `AGENT_VERSIONS.BRIEF_JSON`; * = REQUIRED)

| # | FIELD | TYPE | REQ | VALIDATION / LIMITS | DEFAULT (FROM TEMPLATE) |
|---|---|---|---|---|---|
| 1 | About your business | textarea | ✓ | 20–1,000 chars | placeholder example per industry |
| 2 | What should your agent achieve on a call? | textarea | ✓ | 20–800 chars | template goal text, editable |
| 3 | Business hours | weekly hours editor (Mon–Sun open/close, closed toggle, “same every day” shortcut) | – | valid ranges | Mon–Fri 9–5 |
| 4 | Address / location | text | – | ≤ 200 chars | — |
| 5 | Services / products | tag list (chips) + free text | – | ≤ 30 chips, ≤ 60 chars each | template suggestions |
| 6 | Pricing basics | textarea | – | ≤ 500 chars | — |
| 7 | FAQ (question → answer pairs) | repeatable rows | – | ≤ 20 pairs; Starter: injected into prompt; Growth+: optionally moved to KB | template FAQ starters |
| 8 | Must never do / say | textarea + preset checkboxes (“Never quote prices”, “Never give medical advice”, “Never promise a time”) | – | ≤ 500 chars | template defaults checked |
| 9 | Hand-off rules | rows: condition (text) → action (Transfer to number / Take a message / Read a message); transfer number (E.164) | – | ≤ 5 rules; transfer number validated | template default (“Emergency → transfer”) |
| 10 | Agent name | text | – | 2–30 chars | template default (e.g. “Sophie”) |
| 11 | Personality | 3 sliders (Formal↔Friendly, Brief↔Chatty, Calm↔Energetic) + free text | – | — | template default |
| 12 | Anything else | textarea | – | ≤ 800 chars | — |
| 13 | Should the agent say it’s an AI if asked? | radio: Yes (recommended) / Only if asked / Never claim to be human | ✓ | — | “Only if asked” (always “never claim human” — enforced) |

| ACTION | FRONTEND | BACKEND |
|---|---|---|
| Autosave (debounced 1.5 s on change) | shows “Saved” / “Saving…” | `PATCH /agents/:id/draft {brief_json}` — writes to current draft version, no LLM call |
| ✦ Generate my agent | Disabled until 1 & 2 valid. Shows 3-stage progress (“Reading your brief… Writing the conversation… Setting up analysis”); right panel animates into the Brief card | `POST /agents/:id/generate` → see §13.3. Returns `{version_no, brief_card, greeting, scenarios[], analysis_fields[]}`. p95 < 10 s. On failure: toast “Couldn’t generate — try again” (no version created). |
| Continue | Enabled only after a successful Generate | sets `onboarding_step=3`; → W3 |
| Regenerate (appears after first generate, secondary) | confirm dialog “This rewrites the conversation from your brief; your test feedback is kept” | same endpoint with `mode: regenerate` |
| Advanced ▸ View prompt (link under preview) | opens read-only drawer with the raw generated prompt; “Edit” only inside dashboard D3 | `GET /agents/:id/versions/:v` |

**States:** unsaved-changes guard on Back; generate button shows credits note (“Free — no credits used”). Validation errors scroll to first invalid field.

## W3 — Voice & languages

Route `/onboarding/voice` · Step 3 of 7.

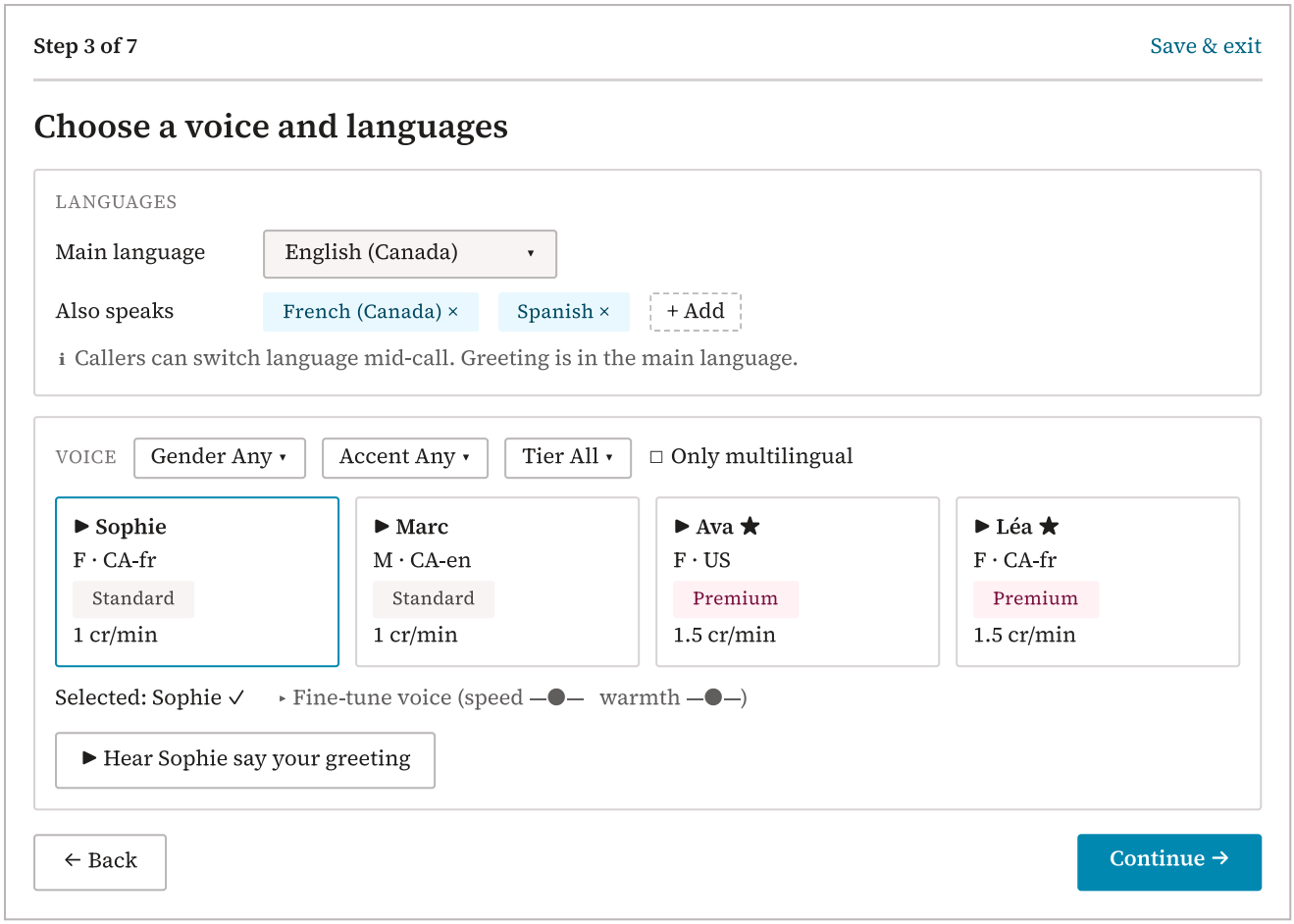


Figure 6 — W3 Voice & languages. Languages set the voice filter; tier and credit rate are shown on every card.

| ELEMENT | TYPE | REQ | VALIDATION | DEFAULT | STORED / MAPPED TO |
|---|---|---|---|---|---|
| Main language | select (locale list Retell supports: en-US, en-CA, en-GB, en-AU, en-IN, fr-FR, fr-CA, es-ES, es-419, de-DE, hi-IN, pt-BR, it-IT, ja-JP, …) | ✓ | — | from workspace country | `agents.primary_language`; Retell agent `language` (or `multi` when >1) |
| Also speaks | multi-select chips | – | ≤ 4 | — | `agents.languages[]`; if ≥ 1 → Retell `language: multi` |
| Voice filters | selects | – | — | gender Any, accent = main language | client-side filter of cached voice list |
| Voice card | radio card; ▶ plays provider preview (from `GET /list-voices` `preview_audio_url`) | ✓ | must support main language; warning if it does not support an “also speaks” language | template default for locale | `agents.voice_id`, `agents.tier` (Premium if ElevenLabs) |
| Speed | slider 0.8–1.2 | – | — | 1.0 | Retell agent `voice_speed` |
| Warmth (voice temperature) | slider 0–1 | – | — | provider default | Retell agent `voice_temperature` |
| Hear my greeting | button | – | rate-limited 10/min | — | `POST /voices/:id/preview` `{text}` → TTS sample via Retell (if available) or cached provider preview; no credits |

| ACTION | FRONTEND | BACKEND |
|---|---|---|
| Change main language | re-filters voices; if selected voice unsupported → auto-pick default and toast | `PATCH /agents/:id/draft {primary_language}`; greeting regenerated in that language by LLM (async, < 3 s) |
| Add “also speaks” | chip added; “Only multilingual” filter auto-enabled | `PATCH …{languages}` |
| Select voice | card highlighted; tier badge and credit rate shown in a confirm strip (“Premium voices use 1.5 credits/min”) | `PATCH …{voice_id, tier}` |
| Continue | — | `PATCH /agents/:id/draft` (final) → pushes voice/language to Retell agent (`PATCH /update-agent`) → `onboarding_step=4` |

**States:** voices cached server-side daily; if the Retell voice API is down, serve cache with a banner. Preview audio plays one at a time.

## W4 — Knowledge (optional)

Route `/onboarding/knowledge` · Step 4 of 7 · Skippable. **Purpose:** give the agent documents/website content. The Starter plan sees the FAQ box instead of a KB.

| ELEMENT | TYPE | REQ | VALIDATION | NOTES |
|---|---|---|---|---|
| Add website URL | url + “Fetch” | – | https, reachable | Retell KB URL source; crawl depth 1 (verify limits) |
| Upload files | dropzone (PDF, DOCX, TXT) | – | ≤ 10 files, ≤ 20 MB each | Retell KB file source |
| Paste text | textarea | – | ≤ 20,000 chars | Retell KB text source |
| Source list | table: name, type, status (Processing/Ready/Failed), size, remove | — | — | — |
| Plan gate (Starter/trial) | card: “Knowledge base is included from the Growth pack. For now, add FAQs in the brief.” + “Add FAQs” link → W2 #7 | — | — | trial users can still **preview** the UI but not add sources |

| ACTION | FRONTEND | BACKEND |
|---|---|---|
| Fetch URL / Upload / Add text | optimistic row “Processing…”, poll status every 3 s | `POST /agents/:id/knowledge {type, url\|file\|text}` → if no KB yet, `POST /create-knowledge-base` on Retell (name = `{tenant}-{agent}`) → add source → job polls Retell until ready → `knowledge_sources.status` |
| Remove | confirm | `DELETE /agents/:id/knowledge/:sourceId` |
| Skip for now / Continue | — | `onboarding_step=5` |

## W5 — Test & improve

Route `/onboarding/test` · Step 5 of 7. **Purpose:** the customer hears their agent, gives feedback in plain English, iterates, and gains confidence before paying.

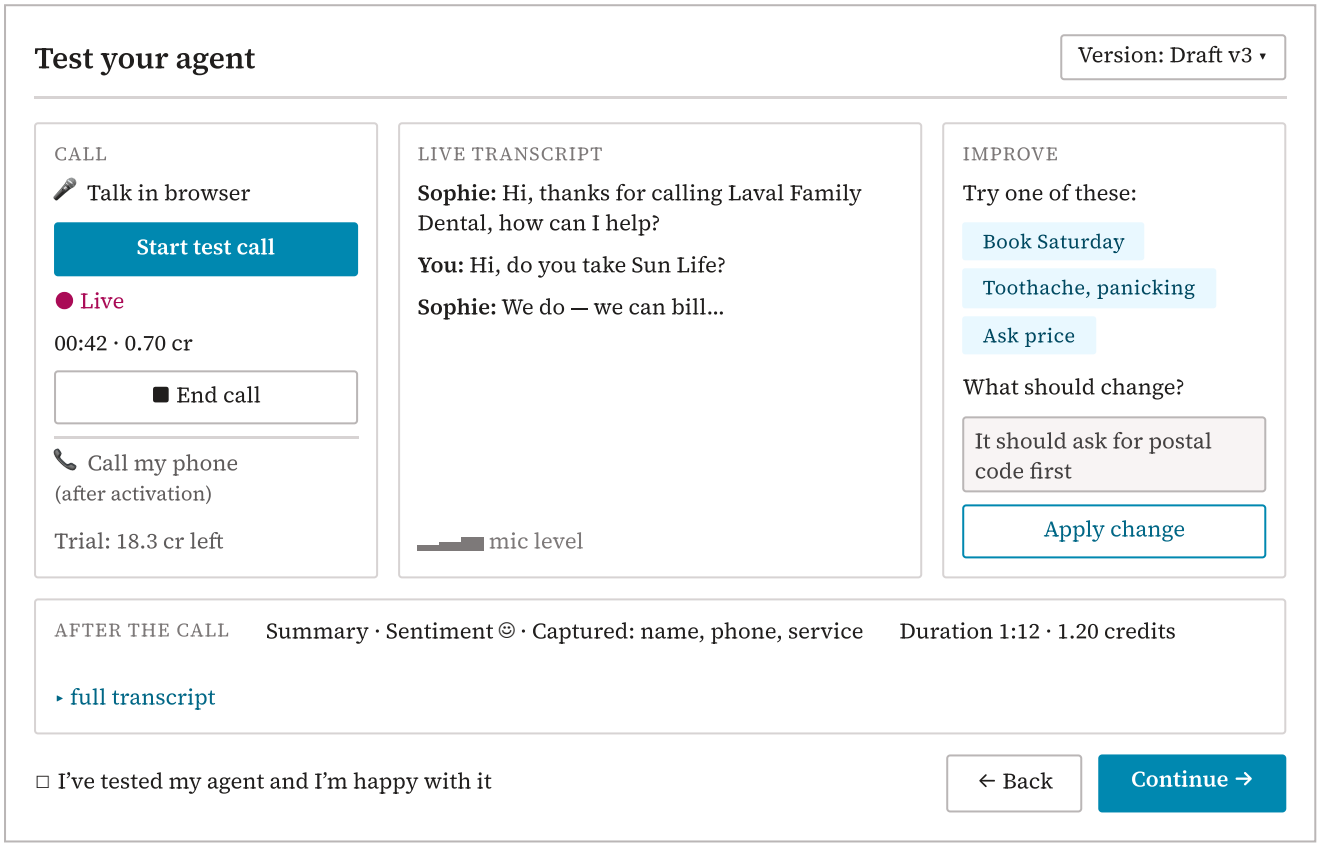


Figure 7 — W5 Test & improve. Call controls, streamed transcript and the plain-language feedback panel side by side; the after-call card fills in when `call_analyzed` arrives.

| ELEMENT | TYPE | NOTES |
|---|---|---|
| Version selector | dropdown of draft versions (v1 generated, v2 feedback, …) | tests run against the selected version; default latest draft |
| Start test call | primary button; requests mic permission | disabled if balance ≤ 0 or unverified email |
| Status / timer / credits | live | credits estimate = elapsed × rate, reconciled on `call_ended` |
| End call | danger secondary | — |
| Call my phone | tel input + button | hidden until first purchase (W7); costs credits; Phase 2 |
| Live transcript | streamed from Retell web SDK `update` events; agent/user bubbles; auto-scroll | — |
| Scenario chips | 3–5 from generation output; clicking shows “Try saying: …” hint | — |
| What should change? | textarea 10–500 chars + **Apply change** | creates a new draft version (§9.6) |
| After-the-call card | summary, sentiment, captured fields, duration, credits, link to transcript | filled when `call_analyzed` arrives (push via SSE, or poll every 3 s up to 60 s) |
| Happiness checkbox | required to Continue (soft gate; “Skip testing” link available with warning) | analytics event |

| ACTION | FRONTEND | BACKEND |
|---|---|---|
| Start test call | `POST /agents/:id/test/web {version_no}` → `{access_token, call_id}` → `retellClient.startCall({accessToken})`; bind SDK events: `call_started`, `update` (transcript), `call_ended`, `error` | Verify agent ∈ workspace; email verified; balance > 0 (trial ok); rate limit 10 test calls/hour/workspace; `POST /v2/create-web-call {agent_id, agent_version, metadata:{tenant_id, agent_id, test:true, version_no}}`; insert `calls` (channel=web, is_test=true, status=registered); return token |
| End call | `retellClient.stopCall()` | Retell fires `call_ended` → metering (§14) |
| Apply change | shows diff card “What changed: • Asks postal code before confirming • …” with **Keep** / **Undo** | `POST /agents/:id/feedback {text, base_version}` → LLM patch → new `agent_versions` row (source=feedback) → `PATCH /update-retell-llm` (draft) → returns `{version_no, changes[]}` |
| Continue | — | `onboarding_step=6` |

**States:** mic denied → inline instructions per browser; Retell SDK error → “Couldn’t connect, try again” + log; balance 0 during trial → “You’ve used your free test minutes — activate to keep testing” with CTA to W6.

## W6 — Phone number

Route `/onboarding/phone` · Step 6 of 7 · Owner/Admin. **Purpose:** decide how calls reach the agent. Nothing is provisioned here — the choice is saved and executed after payment in W7.

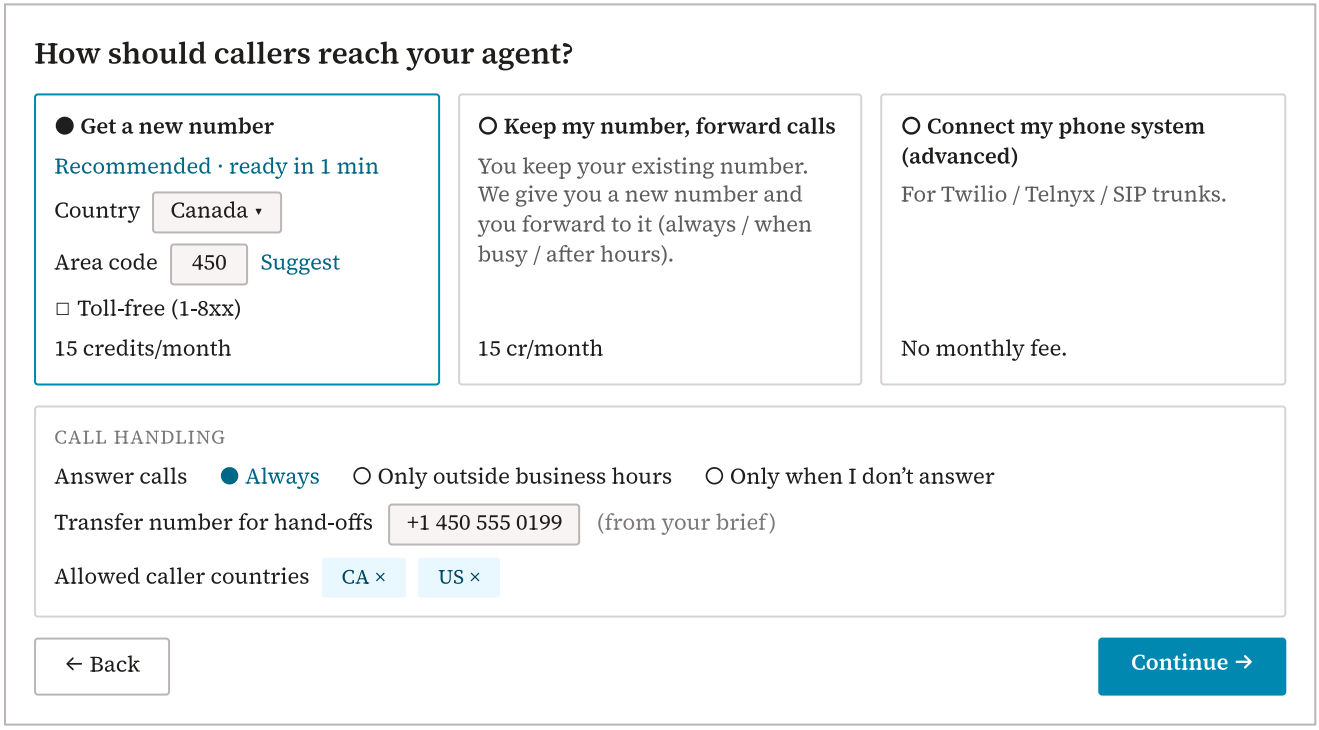


Figure 8 — W6 Phone number. Three routing options as radio cards; call-handling rules apply to whichever is chosen.

OPTION A — GET A NEW NUMBER

| FIELD | TYPE | REQ | VALIDATION | NOTES |
|---|---|---|---|---|
| Country | select (US, CA) | ✓ | — | default workspace country |
| Area code | 3-digit | ✓ (local) | numeric; must exist for country; “Suggest” fills from workspace address/city | Retell assigns a number in that area code; we cannot show candidates before purchase — copy says “We’ll assign you a {area} number” |
| Toll-free | checkbox | – | — | telephony rate to verify (§21); if higher, mark as Premium telephony |

OPTION B — KEEP MY NUMBER, FORWARD CALLS

- Same fields as A (a forward target is still needed) plus **Your current number** (E.164, used in the forwarding guide and to display on the success screen).
- Radio **Forward —** Always / When busy or no answer / Outside business hours (informational; actual forwarding is done on the customer’s carrier).

OPTION C — CONNECT MY PHONE SYSTEM (SIP)

| FIELD | TYPE | REQ | VALIDATION |
|---|---|---|---|
| Phone number | E.164 | ✓ | libphonenumber valid |
| Provider | select: Twilio, Telnyx, Vonage, Other SIP | ✓ | drives the instructions panel |
| Termination SIP URI | text | ✓ | hostname (e.g. `xxx.pstn.twilio.com`) |
| Auth username / password | text / password | – | stored encrypted (KMS) |
| Transport | select TCP/UDP/TLS | ✓ | default TCP |
| “I understand this requires configuring my carrier” | checkbox | ✓ | gate |

CALL HANDLING (ALL OPTIONS)

| FIELD | TYPE | DEFAULT | MAPPED TO |
|---|---|---|---|
| Answer calls | radio | Always | `phone_numbers.hours_mode`; inbound webhook uses workspace hours to set dynamic variable `business_open` and (for "outside hours only") declines during open hours (customer's own phone rings first) |
| Transfer number | E.164 | from brief hand-off rule | `phone_numbers.transfer_targets_json[0]`; Retell transfer tool destination |
| Allowed caller countries | multi-select | workspace country (+US/CA) | Retell `allowed_inbound_country_list` |

| ACTION | FRONTEND | BACKEND |
|---|---|---|
| Continue | validates the selected option's fields | `PUT /agents/:id/phone-plan {option, …fields}` → stored in `phone_plans` (pending) → `onboarding_step=7` |

## W7 — Activate: buy credits & go live

Route `/onboarding/activate` · Step 7 of 7 · Owner/Admin.

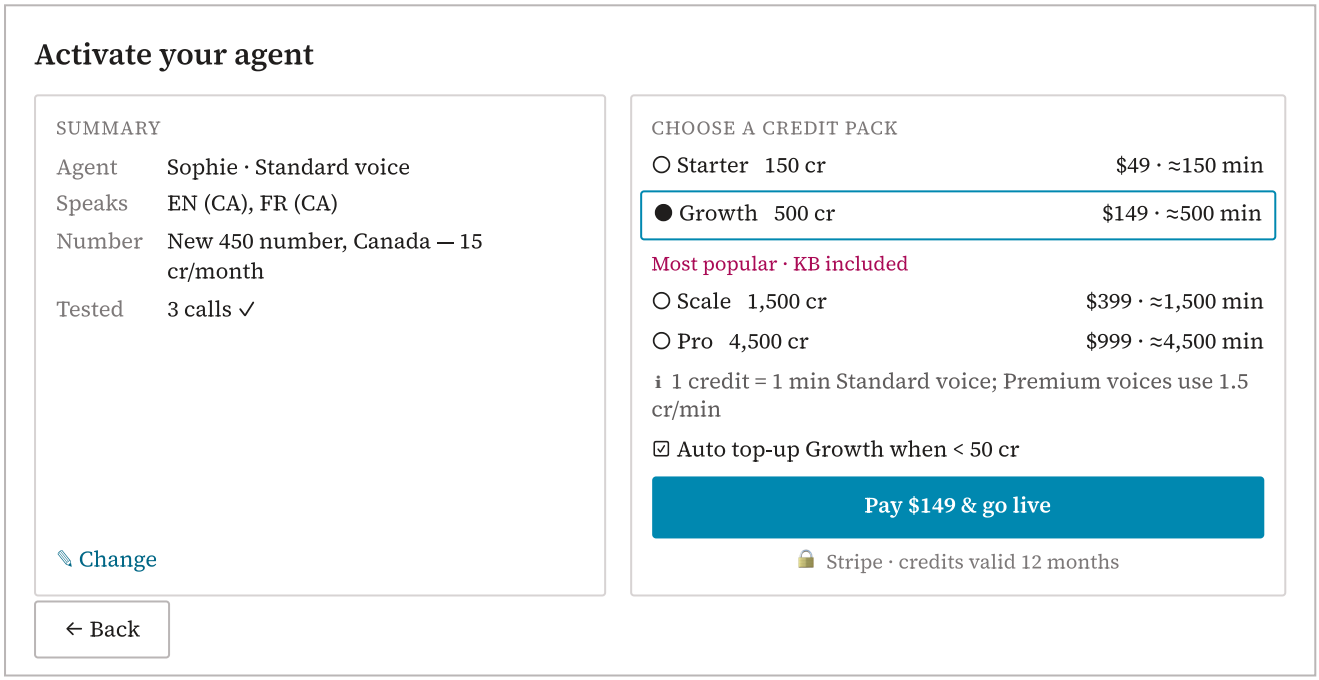


Figure 9 — W7 Activate. Read-only summary left, pack selection and the single paying action right.

| ELEMENT | TYPE | NOTES |
|---|---|---|
| Summary card | read-only with “Change” links back to W3/W6 | — |
| Pack radio cards | single-select | prices/credits from `packs` config table; Growth pre-selected |
| Auto top-up | checkbox + threshold select (10%/20% of pack) | stored after successful payment |
| Pay & go live | primary | label shows amount |

| ACTION | FRONTEND | BACKEND |
|---|---|---|
| Pay & go live | redirect to Stripe Checkout (hosted) | `POST /wallet/checkout {pack_id, auto_topup, agent_id}` → creates `purchases` (pending) → Stripe Checkout Session (`mode=payment`, `setup_future_usage=off_session` if auto top-up) → URL |
| Return from Stripe (`/onboarding/activate?session_id=…`) | shows **Provisioning** progress list: ① Payment confirmed ② Publishing agent ③ Setting up your number ④ Connecting — each ticks as SSE events arrive | Stripe webhook `checkout.session.completed` → `purchases.status=paid` → ledger credit (+pack credits, expires +12 mo) → set plan tier → enqueue `activate-agent` job (§13.8): publish agent version → provision/import number → bind number → mark live → emit SSE |
| Failure at any step | progress row shows ✖ with “Retry” (job re-run) and support link; credits are never rolled back automatically | job is idempotent; number provisioning failure alerts ops |

### W7b — You’re live (success)

Route `/onboarding/live`.

- Big number display “+1 (450) 555-0123” with **Copy** and **Call it now** (tel: link on mobile).
- If Option B: forwarding guide panel (carrier select → exact dial codes, e.g. `*72` + number; “forward when busy” `*90`/`*92`; screenshots for major carriers; “I’ve set up forwarding” checkbox).
- If Option C: SIP origination instructions with the Retell SIP URI and a “Test inbound” hint.
- Next-steps checklist: invite a teammate, turn on daily digest, add your website widget (Phase 2).
- Button **Go to dashboard** → `/app`; `tenants.onboarding_completed_at` set.

# 09 Prompt system and template library

## 9.1 Principles

- The customer edits a **Brief** (structured fields). We generate the **prompt** (Retell LLM `general_prompt`), the **greeting** (`begin_message`), the **post-call analysis schema**, and **test scenarios**.
- Fixed **platform guardrails** are appended to every prompt and cannot be removed by the customer.
- Every generation or edit creates an immutable **version**. Numbers always run the **published** version.
- The raw prompt is editable only in the dashboard under **Advanced** (D3 ▸ Brief ▸ Advanced), with lint warnings.

## 9.2 Template catalogue (MVP)

### AI Receptionist `receptionist` Standard / locale default

| | |
|---|---|
| BEST FOR | any service business |
| PRE-FILLED GOAL | “Answer calls, share hours/location/services, take detailed messages, transfer urgent calls” |
| STEPS | 1 greet · 2 identify need · 3 answer from business facts/FAQ · 4 if can’t → take message (name, number, reason) · 5 confirm details · 6 close |
| CAPTURED | caller_name (text), callback_number (text), reason (text), urgency (enum low/medium/high), message_taken (bool) |
| DEFAULT RULES | never quote prices unless in brief; never promise a callback time; say “I’m an AI assistant” if asked |

### Appointment Request `appointment` Standard

| | |
|---|---|
| BEST FOR | clinics, salons, trades |
| PRE-FILLED GOAL | “Collect appointment requests with preferred times so the team can confirm” |
| STEPS | 1 greet · 2 which service · 3 new or existing client · 4 preferred days/times (collect 2 options) · 5 name, phone · 6 confirm & explain a person will confirm · 7 close |
| CAPTURED | service (text), preferred_slots (text), is_new_client (bool), caller_name, callback_number |
| DEFAULT RULES | never confirm a slot as booked; ask 2 alternatives |

### After-hours Answering `afterhours` Standard

| | |
|---|---|
| BEST FOR | any, forwarded lines |
| PRE-FILLED GOAL | “Handle calls when closed, collect messages, escalate emergencies” |
| STEPS | 1 greet + state closed & reopening time () · 2 is this urgent? · 3 if emergency per rules → transfer/read emergency line · 4 else take message · 5 close |
| CAPTURED | is_emergency (bool), caller_name, callback_number, message (text) |
| DEFAULT RULES | emergency definitions from brief; never give safety/medical advice |

### Lead Qualification `lead_qual` Premium suggested

| | |
|---|---|
| BEST FOR | real estate, agencies, B2B services |
| PRE-FILLED GOAL | “Qualify inbound leads and capture contact details” |
| STEPS | 1 greet · 2 what are they looking for · 3 3–5 qualifying questions (from brief: budget, timeline, location, decision-maker) · 4 contact details · 5 next step promise · 6 close |
| CAPTURED | name, phone, email (text), need (text), budget (text), timeline (text), qualified (bool), score (number 1–5) |
| DEFAULT RULES | never quote final pricing; don’t push if not interested |

### FAQ / Support `faq_support` Standard

| | |
|---|---|
| BEST FOR | product/support lines |
| PRE-FILLED GOAL | “Answer questions from the knowledge base; escalate what you can’t answer” |
| STEPS | 1 greet · 2 question · 3 answer from KB with citation-style honesty · 4 if unknown → offer message/transfer · 5 close |
| CAPTURED | topic (text), resolved (bool), follow_up_required (bool), callback_number |
| DEFAULT RULES | never invent answers; say “I don’t have that information” |

### Restaurant Reservations `restaurant` Standard

| | |
|---|---|
| BEST FOR | restaurants, cafés |
| PRE-FILLED GOAL | “Take reservation requests, answer hours/menu/parking, hand off large parties” |
| STEPS | 1 greet · 2 reservation or question · 3 party size, date, time, name, phone · 4 large party (> N) → transfer/message · 5 confirm · 6 close |
| CAPTURED | party_size (number), date_time (text), name, phone, special_requests (text) |
| DEFAULT RULES | never confirm availability as guaranteed |

### Custom `custom` Standard

| | |
|---|---|
| BEST FOR | anything |
| PRE-FILLED GOAL | blank |
| STEPS | generic 1 greet · 2 identify · 3 help · 4 capture · 5 close |
| CAPTURED | caller_name, callback_number, reason |
| DEFAULT RULES | platform rules only |

Templates live in a versioned config file (`templates/*.json`) with: brief prefill, conversation steps, analysis fields, rules, sample call audio path, industry-specific example texts (dental, plumbing, legal, salon, auto, real estate, restaurant, generic).

## 9.3 Generation pipeline (Brief → agent)

1. **Frontend** — `POST /agents/:id/generate`
2. **Backend builds the LLM request** (Anthropic API, Claude Sonnet class)
   *system:* role ("You write production voice-agent prompts for Retell…"), output JSON schema, style rules (short sentences, spoken-language, no markdown in spoken lines, ≤ 6,000 chars prompt)
   *user:* template skeleton (Appendix B) + `brief_json` + locale + tier
3. **LLM returns JSON** — `{prompt, greeting, closing, brief_card:{does[], doesnt[], handoff[]}, analysis_fields:[{name,type,description,enum?}], scenarios:[{title, say_this}], warnings[]}`
4. **Validate** with zod → append platform guardrails block → run lint (§9.5)
5. **Persist** `agent_versions` (source=`generated`, `prompt_text`, `greeting`, `analysis_schema_json`, `brief_card_json`, `scenarios_json`)
6. **Push to Retell** — `POST /create-retell-llm` (first time) or `PATCH /update-retell-llm {general_prompt, begin_message, model, general_tools:[transfer_call…]}`; `PATCH /update-agent {response_engine, voice_id, language, post_call_analysis_data, webhook_url}`. *Draft only — not published.*
7. **Return summary** to frontend. Target p95 ≤ 10 s; jobs > 15 s time out with retry.

Figure 10 — Generation pipeline. Step 6 (magenta) is the only step that touches Retell; everything before it is reversible.

## 9.4 Platform guardrails (appended, non-editable)

- Never claim to be human; if asked, say you are an AI assistant for {business}.
- Never provide medical, legal or financial advice beyond facts in the brief/KB; recommend speaking to the business.
- Never invent prices, availability, policies or names. If unknown: "I don't have that information, but I can take a message."
- Never collect payment card numbers, SSN/SIN, passwords or full health details; ask for a callback instead.
- No profanity, no political/religious opinions, de-escalate abuse once then end politely.
- Keep turns ≤ 2 sentences; confirm captured details back; end the call when the caller is done.
- Respect : when closed, say so and follow after-hours rules.

## 9.5 Prompt lint (warnings shown in Advanced editor and on Publish)

Length > 6,000 chars (latency/cost) · missing hand-off rule when transfer number set · transfer number set but no transfer tool · no greeting · greeting > 25 words · "always"/"guarantee" language in pricing context · language mismatch (prompt language ≠ primary language).

## 9.6 Feedback loop (plain-language improvement)

- Input: free text (10–500 chars) + base version.
- LLM receives current brief + prompt + feedback; returns `{updated_brief_json (only changed fields), updated_prompt, changes:[human-readable bullets]}`.
- New version created with `source=feedback`, `feedback_text` stored for audit. UI shows bullets with Keep/Undo; Undo simply reselects the previous version as draft.

- Guardrails are re-appended; lint re-run; feedback that conflicts with a guardrail returns `changes: []` + explanation ("I can't make the agent claim to be human").

### 9.7 Dynamic variables available in prompts

, , , (workspace TZ), (true/false), , . Set per call by the inbound webhook (§13.9); for web tests, computed at token creation.

### 9.8 Post-call analysis schema

- Fields from the template + user-added (Growth+): name (snake_case), type (text/number/boolean/enum), description, enum values. Max 15 fields.
- Mapped to Retell `post_call_analysis_data`. Results arrive in `call_analyzed` → `calls.analysis_json` → shown in D2 and in the daily digest.

### 9.9 Versioning and publishing

- `agent_versions.version_no` increments per agent; `source ∈ {generated, feedback, manual, regenerate}`.
- **Publish** = `POST /agents/:id/publish {version_no}` → Retell publish agent version (or create version & tag) → `agents.published_version` → numbers bound to `latest_published` pick it up on the next call. Old published versions remain restorable ("Roll back" = publish an older version).
- Drafts are tested with explicit `agent_version` in `create-web-call`; production calls never use drafts.

# 10 Dashboard — screen-by-screen

All dashboard screens use the global shell (§7.2). Default landing after login is D1.

### D1 — Home

Route `/app` · All roles.

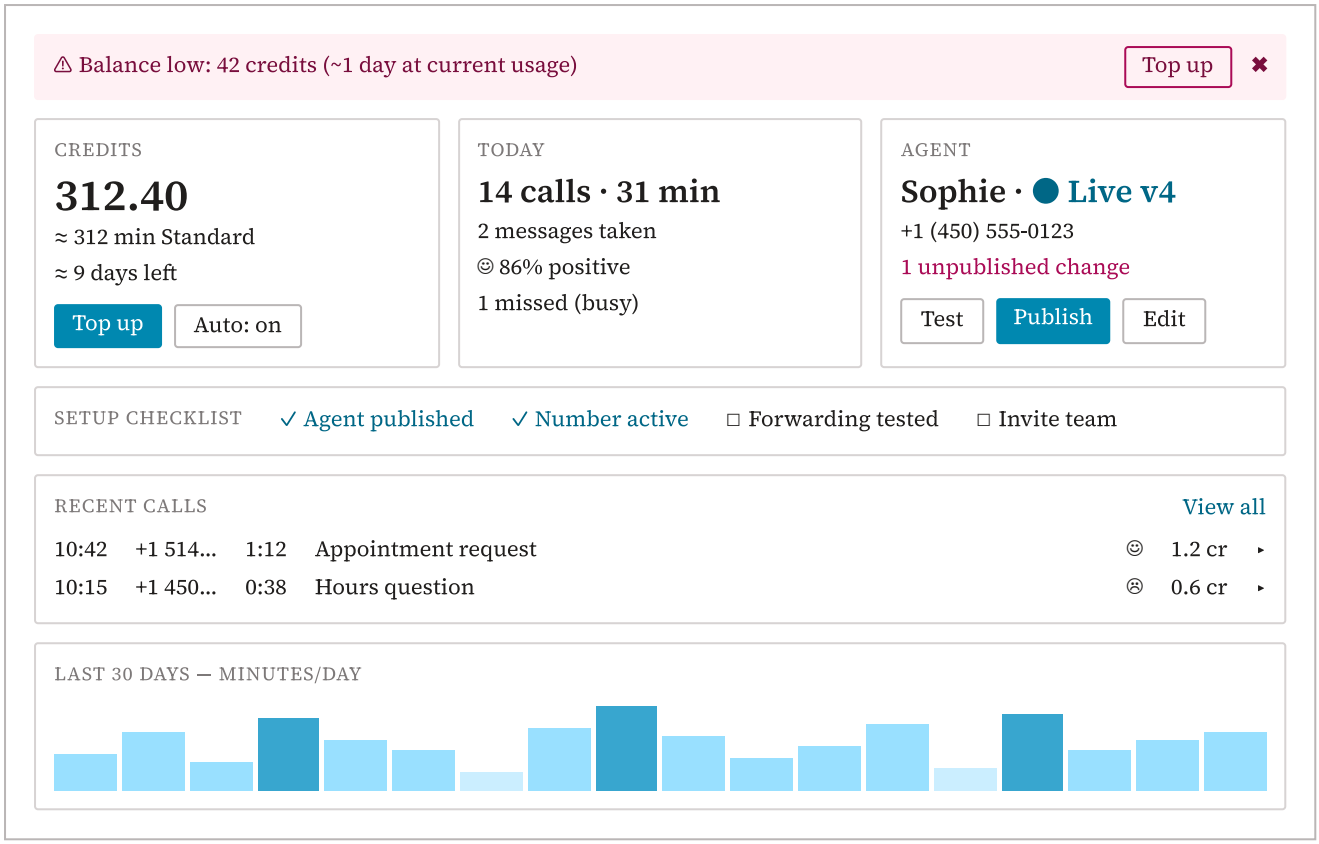


Figure 11 — D1 Home. Alert banner, three status cards, setup checklist, recent calls and the 30-day usage chart.

| WIDGET | DATA | API |
|---|---|---|
| Alert banner | low balance / paused / failed top-up / number issue | `GET /alerts/active` |
| Credits card | balance, ≈ minutes at Standard, days left = balance ÷ 7-day avg daily burn, buttons Top up (→ D5), Auto top-up state | `GET /wallet` |
| Today card | calls, minutes, messages (analysis `message_taken` or `callback_number` present), sentiment %, missed (declined at cap/balance) | `GET /stats?range=today` |
| Agent card | name, status (Live/Draft/Paused), published version, number, unpublished-changes chip; Test → D3 test tab; Publish → confirm modal → `POST /agents/:id/publish`; Edit → D3 brief | `GET /agents` |
| Setup checklist | derived flags | `GET /workspace/checklist` |
| Recent calls | last 5 | `GET /calls?limit=5` |
| 30-day chart | minutes per day | `GET /stats?range=30d` |

**States:** a new workspace without a published agent gets a “Finish setup” hero with a resume button; Viewer sees no Top up/Publish buttons.

## D2 — Calls

Route `/app/calls` · All roles (recordings/transcripts gated by role? all roles can view in MVP).

LIST

| COLUMN | SOURCE |
|---|---|
| Date/time (workspace TZ) | `calls.started_at` |
| Caller (formatted, country flag) | `calls.from_number` |
| Number / agent | join |
| Duration | `duration_s` |
| Outcome chip | derived: Message taken / Appointment request / Transferred / Resolved / Hung up early (< 10 s) / Missed (busy/paused) |
| Sentiment | `analysis_json.user_sentiment` (☺ 😐 ☹) |
| Credits | `credits_charged` |
| Status dot | Handled / Needs attention (toggle) |

Filters (URL-synced): date range (Today/7d/30d/custom), agent, number, direction (Phase 2), outcome, sentiment, needs-attention, free-text search over transcript (Postgres full-text) and caller number. Sort by time/duration/credits. Pagination 25/page. **Export CSV** (current filter, max 10k rows, async email if > 2k). Bulk: mark handled.

DETAIL (DRAWER ON DESKTOP, PAGE ON MOBILE) `/APP/CALLS/[ID]`

| SECTION | CONTENT | ACTIONS |
|---|---|---|
| Header | caller, time, duration, credits, agent + version, number, outcome | Mark handled/unhandled · Add note (≤ 500 chars) · Copy caller number · **Call back** (tel: link; Phase 2 click-to-call) |
| Summary & data | LLM summary, sentiment, call_successful, each captured field as label/value, "follow up required" | Copy all |
| Transcript | bubbles with timestamps; agent/caller; search-highlight; tool-call markers (transfer) | Copy transcript · Report a problem (creates support ticket with call id) |
| Recording | audio player (Retell recording URL proxied via our backend, signed, 1-h TTL) | Download (Owner/Admin) |
| Details | call_id, disconnection reason (plain-English mapped), version, tier, Retell cost (staff only) | — |

**Empty state:** "No calls yet — call your number to try it". **Loading:** skeleton rows. **Real-time:** new-call toast via SSE (`/events`) while the list is open.

## D3 — Agents

Route `/app/agents` (list) · `/app/agents/[id]/*` (detail tabs). Editor+ can edit; Viewer read-only.

**List columns:** name, status (Live/Draft only/Paused), tier badge, languages, numbers attached, calls (7d), avg duration, last published, ⋯ menu (Duplicate, Pause/Resume, Delete — Owner/Admin). **+ New agent** (Growth+: multiple agents; Starter: 1) → launches the wizard in "agent mode" (W1–W5 then attach to number in D4).

| TAB | CONTENT | ACTIONS |
|---|---|---|
| Overview | Brief card (does/doesn't/hand-off), version pill (Draft v5 · Published v4), attached numbers, 7-day stats, quick Test button | Publish / Discard draft |
| Brief | Same form as W2 (all 13 fields) with autosave; right panel Brief card; **Regenerate**; **Advanced ▸ Edit prompt** (raw textarea, lint panel, "Reset to generated", character counter); **Analysis fields** editor (Growth+) | Save creates version (source=manual) on explicit **Save as new version** |
| Voice & language | Same as W3 | autosave to draft |
| Knowledge | Same as W4 + usage stats (KB hits per call, Phase 2) | — |
| Test & improve | Same as W5 plus **Call my phone** (Phase 2) and test-call history (test calls flagged, excluded from analytics) | — |
| Versions | table: v#, source, author, created, notes/changes bullets, Published badge; **Preview** (read-only prompt diff vs published), **Test this version**, **Publish**, **Restore as draft** | Publish confirm modal lists lint warnings and "This affects live calls immediately" |
| Settings | Agent name; Tier (Standard/Premium — shows credit rate); Recording on/off; Max call duration (default 15 min, 3–30); Voicemail detection (Phase 2); Denoising add-on (+0.005 cost absorbed, toggle Scale+); PII removal (Scale+); Delete agent | — |

Sticky bottom bar when draft ≠ published: "Unpublished changes · [Test] [Publish]".

## D4 — Numbers

Route `/app/numbers` · Owner/Admin edit; others view.

**List:** number (formatted), label/nickname, type (New / Forward target / SIP), country, attached agent, status (Active/Provisioning/Paused/Releasing), calls 7d, rental (15 cr/mo or —), ⋯ (Rename, Change agent, Release). **+ Add number** → modal with the W6 options (buy immediately if balance ≥ 15 credits; SIP import).

| SECTION | FIELDS / CONTROLS | BACKEND |
|---|---|---|
| Routing | Attached agent (select, published only) · Answer calls: Always / Outside hours / When I don't answer · Business hours (uses workspace hours; "override for this number" toggle → weekly editor) | `PATCH /numbers/:id` → `PATCH /update-phone-number {inbound_agents}`; hours stored locally, applied by inbound webhook |
| Hand-off | up to 3 transfer targets (label, E.164, when: rule text) · Read-out message if transfer fails | stored in `transfer_targets_json`; injected as dynamic variables + transfer tool config on the Retell LLM |
| Callers | Allowed caller countries (multi) · Block list (E.164 list, ≤ 100) | `allowed_inbound_country_list`; block list enforced in inbound webhook (decline) |
| Forwarding guide (Forward-target type) | carrier select → codes; "Test forwarding" (Phase 2: we call their original number and verify it lands) | — |
| SIP (SIP type) | termination URI, username (masked), transport, Retell origination URI to configure on their trunk; "Re-verify" | `PATCH /update-phone-number` |
| Danger zone | Release number (confirm with number typed; releases in Retell, stops rental; 24-h undo window not possible — warn) | `DELETE /delete-phone-number` |

## D5 — Wallet & billing

Route `/app/wallet` · Owner/Admin full; Editor/Viewer see balance + ledger only.

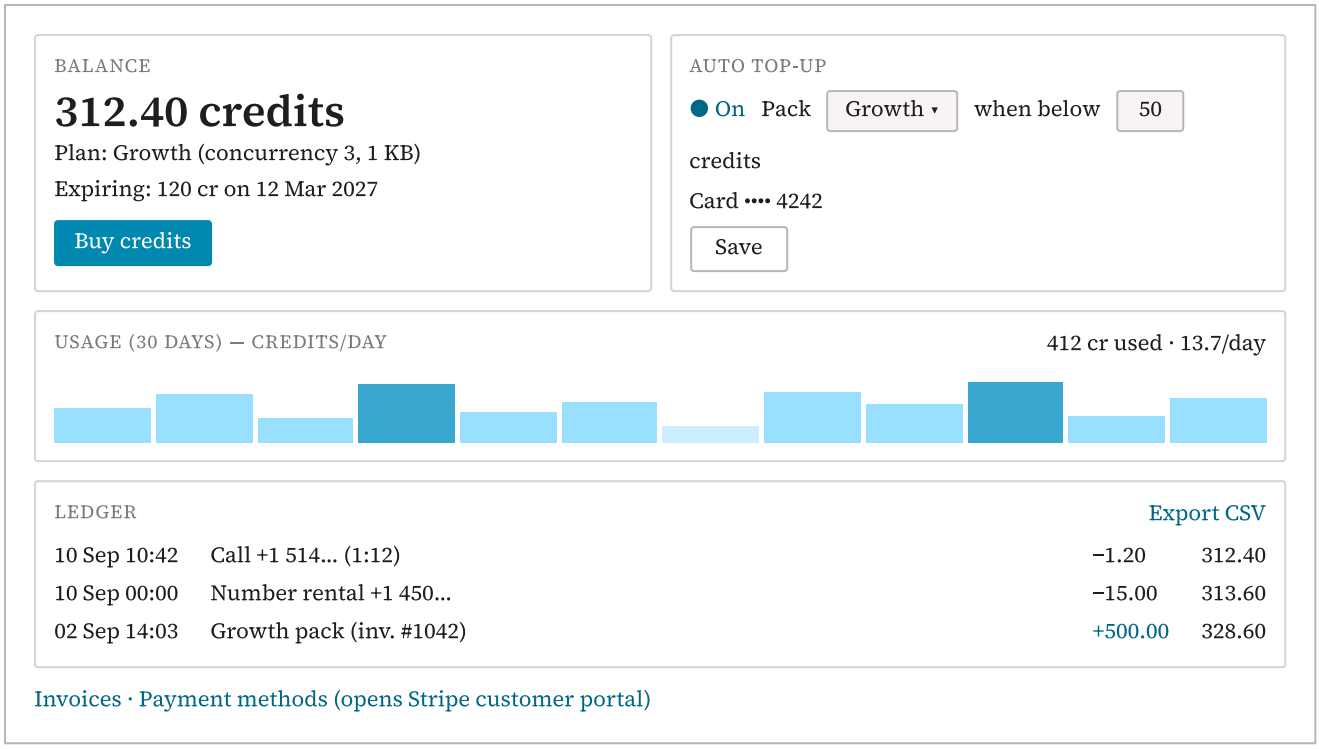


Figure 12 — D5 Wallet. Balance and auto top-up side by side, usage chart, append-only ledger.

| ELEMENT | BEHAVIOUR | BACKEND |
|---|---|---|
| Buy credits | modal with pack cards (same as W7) → Stripe Checkout | `POST /wallet/checkout` |
| Auto top-up | on/off, pack, threshold (numeric ≥ 10), card from Stripe (must exist; else “Add card” → Stripe portal) | `PUT /wallet/auto-topup`; requires MFA if enabling (§6.2) |
| Expiring soon | lots expiring within 60 days | `GET /wallet/lots` |
| Ledger | paginated; filters by reason; each call row links to D2 detail | `GET /wallet/ledger` |
| Invoices | Stripe-hosted invoice/receipt links | `GET /wallet/invoices` |
| Payment methods | Stripe Customer Portal (cards, billing address, tax ID) | `POST /wallet/portal-session` |

## D6 — Settings

Route `/app/settings/*`.

| SUB-PAGE | FIELDS / ACTIONS | ROLES |
|---|---|---|
| Workspace | Business name, country (locked after first number), time zone, address (used for consent rules + area-code suggest), default business hours (weekly editor; used by all numbers unless overridden), website | Owner/Admin |
| Team | member table (name, email, role, last active, MFA ✓), Invite (email + role) → `POST /invites`, change role, remove, transfer ownership (Owner only; email confirmation) | Owner/Admin |
| Notifications | per-user: email on new message captured (instant/daily digest), low balance (email/SMS), weekly summary; workspace-level: digest recipients | all (own), Owner/Admin (workspace) |
| Security | change password, MFA enrol/disable (with re-auth), active sessions list + revoke, connected SSO accounts | all (own) |
| Data & privacy | recording default for new agents, retention (30/90/365 days) for recordings & transcripts, PII removal default (Scale+), consent line auto-insert (on when two-party jurisdiction), export all data (async zip), delete workspace (Owner; typed confirmation; releases numbers, forfeits credits — explicit warning) | Owner/Admin |
| Audit log | table of events (§6.2) with actor, IP, time; export | Owner/Admin |

### D7 — Help

Docs links (forwarding guides, SIP guides, billing FAQ), in-app support chat (Intercom/Crisp), Retell status embed, “Report a call” shortcut.

### D8 — Internal admin console (`/admin`, staff only)

- Tenants list (plan, balance, burn, Retell agent ids), impersonate (read-only, audited), manual credit adjustment with reason, refund flag.
- Platform metrics: concurrency (current/peak per hour vs Retell allowance), KB count vs free tier, Retell spend vs credit revenue (from `call_cost`), failed webhooks, provisioning failures, reconciliation drift.
- Kill-switches: pause tenant, pause number, global “reject new calls” (incident).

# 11 Phone number configuration (reference)

## 11.1 Option A — Buy a new number (through us)

- Country (US, CA at launch), area-code, local or toll-free.
- Backend: `POST /create-phone-number` with `number_provider` (twilio/telnyx), `area_code`, `inbound_agents: [{agent_id, agent_version: "latest_published"}]`, `inbound_webhook_url` (our per-number endpoint with a signed token), `allowed_inbound_country_list` per plan.

- 15 credits/month rental deducted; number released to Retell if balance stays ≤ 0 for 30 days (warnings at 7, 14, 21 days).
- Toll-free telephony rate on Retell may differ from local — verify (§21); if higher, mark toll-free as Premium telephony (1.25 credits/min) or restrict to Scale+.

### 11.2 Option B — Keep your existing number, forward calls (recommended for most SMBs)

- Customer buys a number through us (Option A) and forwards their existing line to it. No SIP, no porting, reversible in seconds.
- Portal shows carrier-specific instructions: always forward, forward when busy / no answer (overflow), forward outside hours (via their phone system).
- Caller ID: forwarded calls arrive with the original caller's number on most carriers; surfaced in the call log. Some carriers pass the forwarding number instead — documented in the guide.

### 11.3 Option C — Connect your own phone system (SIP, advanced)

- Twilio / Telnyx / Vonage / generic SIP trunk. `POST /import-phone-number` with termination URI + credentials + `inbound_agents` + `inbound_webhook_url`.
- Origination side (their trunk → Retell SIP) documented per provider.
- Telephony cost from Retell = $0 → identical credit rate in MVP; candidate for a lower rate later.

### 11.4 Option D — Port your number (Phase 3)

LOA, carrier coordination, 2–4 weeks. Manual ops process.

### 11.5 Web widget channel (Phase 2)

Embeddable "Talk to us" button using Retell's web-call SDK. No telephony cost. Separate channel in analytics.

# 12 Frontend architecture

### 12.1 Stack

- **Next.js (App Router), TypeScript, Tailwind, shadcn/ui.** Route groups: `(auth)`, `(onboarding)`, `(app)`, `(admin)`.
- **Auth:** provider SDK (Clerk) components for sign-up/login/MFA; our own screens for workspace setup and invites.
- **Data:** TanStack Query for server state (cache keys per workspace); Zustand for wizard/UI state; **React Hook Form + zod** for forms. Zod schemas are shared with the backend package (`@portal/schemas`) so validation is identical.

- **Realtime:** Retell `retell-client-js-sdk` for browser test calls (mic capture, transcript `update` events); Server-Sent Events (`GET /events`) for provisioning progress, new-call toasts, balance updates.
- **i18n:** next-intl, `en` and `fr-CA` at launch; all copy in message files; dates/numbers via Intl.
- **Analytics:** PostHog events per wizard step (`onboarding_step_viewed/completed`, `generate_clicked`, `test_call_started/ended`, `pack_selected`, `checkout_completed`) → funnel dashboard.
- **Error handling:** global error boundary; API errors mapped to inline field errors (422 with `{field, message}`) or toasts; retry with backoff for GETs.
- **Feature flags:** simple `flags` endpoint (plan gates + kill-switches) cached per session.

## 12.2 Wizard state machine

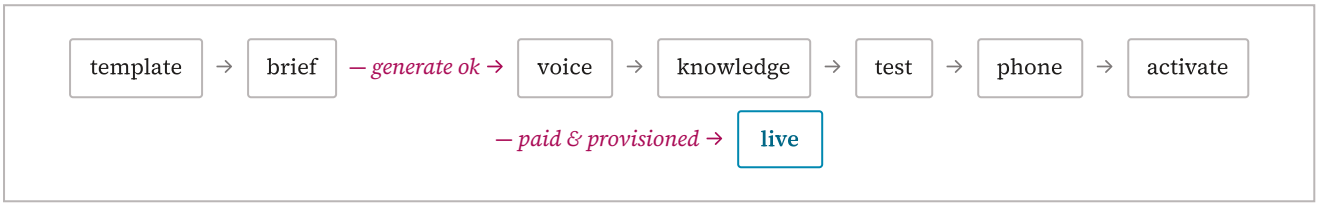


Figure 13 — Wizard state machine. Two gated transitions (magenta): a successful generation, and a completed payment plus provisioning.

- Each step's **Continue** calls a `PATCH/POST` and only advances on 2xx. `tenants.onboarding_step` is the source of truth for resume; the stepper allows jumping back to any completed step.
- Generation and provisioning are **async jobs**; the UI subscribes to SSE topic `agent:{id}` / `activation:{id}` and falls back to polling `GET /jobs/:id` every 3 s.

## 12.3 Test-call client flow

1. `POST /agents/:id/test/web` → `{access_token, call_id}`.
2. `client.startCall({accessToken, sampleRate: 24000})`.
3. Events: `call_started` → status Live, timer starts; `update` → append/replace last transcript turn; `agent_start_talking/stop_talking` → speaking indicator; `call_ended` → status Ended, wait for SSE `call:{id}:analyzed` (≤ 60 s) to show summary; `error` → show retry.
4. `stopCall()` on the End button or route change (prompt "End your test call?").

## 12.4 Component inventory (reused across wizard and dashboard)

`BriefForm` · `BriefCard` · `HoursEditor` · `VoicePicker` · `LanguagePicker` · `KnowledgeSources` · `TestCallPanel` · `TranscriptView` · `ImprovePanel` · `PhoneOptionCards` · `PackSelector` · `ProvisioningProgress` · `BalancePill` · `CallsTable` · `CallDetail` · `LedgerTable` · `VersionList` · `TransferTargetsEditor` · `CarrierForwardingGuide`

### 12.5 Responsiveness

Wizard: single-column on mobile, preview panels become collapsible sections; test call works on mobile Safari/Chrome (mic permission copy per browser). Dashboard: tables become cards on < 640 px.

# 13 Backend architecture and key flows

### 13.1 Stack

- **NestJS (TypeScript) —** matches the shared zod schemas; or FastAPI if the team standard is Python (decision §20). Modules: `auth`, `workspaces`, `agents`, `prompting`, `voices`, `knowledge`, `numbers`, `calls`, `wallet`, `billing`, `webhooks`, `notifications`, `events` (SSE), `jobs`, `admin`.
- **Postgres** (Prisma), **Redis** (balances cache, active-call counters, rate limits, SSE fan-out), **BullMQ** (jobs), **Stripe**, **Retell SDK**, **Anthropic SDK**, **S3-compatible storage** (exports; recordings stay on Retell URLs unless retention policy requires copy).
- `RetellGateway` service is the **only** module that talks to Retell; it owns tier → model/voice mapping, webhook URL construction, and retry/backoff.
- All webhook handlers are **idempotent** and enqueue work; the HTTP handler only validates, stores the raw event (`webhook_events`) and returns 2xx quickly.

### 13.2 Flow — Sign-up and workspace creation

1. Auth provider creates user → webhook `user.created` → `users` upsert.
2. `POST /workspaces` → transaction: `tenants` row (plan `trial`), `memberships` (Owner), `credit_ledger` +20 (reason `trial`, expires +30 days), default `notification_settings`, `audit_log`.
3. Response includes workspace id; frontend stores it and redirects to W1.

### 13.3 Flow — Generate agent (W2)

1. `POST /agents/:id/generate` → authz (Editor+) → load draft brief → enqueue `generate-agent` job → return `{job_id}` (202).
2. Job: build LLM request (§9.3) → call Anthropic (timeout 20 s, 1 retry) → validate JSON → lint → create `agent_versions` (n+1) → `RetellGateway.upsertLlm(agent, version)` → `RetellGateway.upsertAgent(agent, version)` → mark job done → SSE `agent:{id}` `{event: generated, version_no, brief_card, scenarios}`.
3. Retell objects: one `retell_llm_id` and one `retell_agent_id` per agent, created lazily on first generate; later versions `PATCH` the same objects (draft) — publishing creates a Retell agent version.

### 13.4 Flow — Browser test call (W5)

1. `POST /agents/:id/test/web {version_no}` → authz; email verified; `balance > 0`; rate-limit `test:{ws}` 10/h.

2. Compute dynamic variables (`business_open` etc.) → `POST /v2/create-web-call {agent_id, agent_version, retell_llm_dynamic_variables, metadata:{tenant_id, agent_id, version_no, test:true}}`.
3. Insert `calls` (channel `web`, `is_test=true`, status `registered`) → return `{access_token, call_id}`.
4. Retell webhooks `call_started/ended/analyzed` → §14 → SSE `call:{id}` updates.

## 13.5 Flow — Feedback patch (W5 Improve)

1. `POST /agents/:id/feedback {text, base_version}` → job `patch-agent` → LLM (§9.6) → new version (`source=feedback`) → Retell draft `PATCH` → SSE `{event: patched, version_no, changes[]}`.
2. **Undo** = `POST /agents/:id/draft/select {version_no: previous}` (no LLM call).

## 13.6 Flow — Publish

`POST /agents/:id/publish {version_no}` → lint (block on errors, warn on warnings) → `RetellGateway.publish(agent, version)` (Retell publish / version create) → `agents.published_version` → `audit_log` → SSE. Numbers bound to `latest_published` pick it up immediately.

## 13.7 Flow — Checkout (W7 / D5)

1. `POST /wallet/checkout {pack_id, auto_topup?, agent_id?}` → `purchases` (pending) → Stripe Checkout Session with `client_reference_id = purchase_id`, `metadata {workspace_id}`, `setup_future_usage` when auto top-up requested → return URL.
2. Stripe webhook `checkout.session.completed` (verified signature, idempotent on `event.id`) → mark paid → `credit_lots` + `credit_ledger` (+credits, `expires_at`=+12 mo) → update plan tier (max pack in 12 mo) → save payment method id if present → if `agent_id` present enqueue `activate-agent` → email receipt → SSE `wallet` update.
3. `checkout.session.expired` → purchase `abandoned`.

## 13.8 Flow — Activation job (after first payment)

Idempotent steps, each recorded in `activation_jobs.steps_json`:

1. **Publish** latest draft (13.6) if no published version.
2. **Provision number** per `phone_plans`: A/B → `POST /create-phone-number` (area code, provider, country, toll_free, `inbound_agents`, `inbound_webhook_url`); C → `POST /import-phone-number`. Store `phone_numbers` (status `active`), schedule rental (`next_rental_at` = +1 month; first month charged now: −15 credits).
3. **Bind** → confirm `inbound_agents` = published agent; set `allowed_inbound_country_list`.
4. **Mark live** → `agents.status=live`, `tenants.onboarding_completed_at`, checklist flags.
5. **Notify** → email with number + forwarding guide; SSE per step. Failures: retry 3× with backoff; then `failed` with ops alert (Slack) and user-facing Retry.

## 13.9 Flow — Inbound call authorisation (Retell inbound webhook)

`POST /webhooks/retell/inbound/:number_token` (token maps to `phone_numbers.id`; also verify `X-Retell-Signature`).

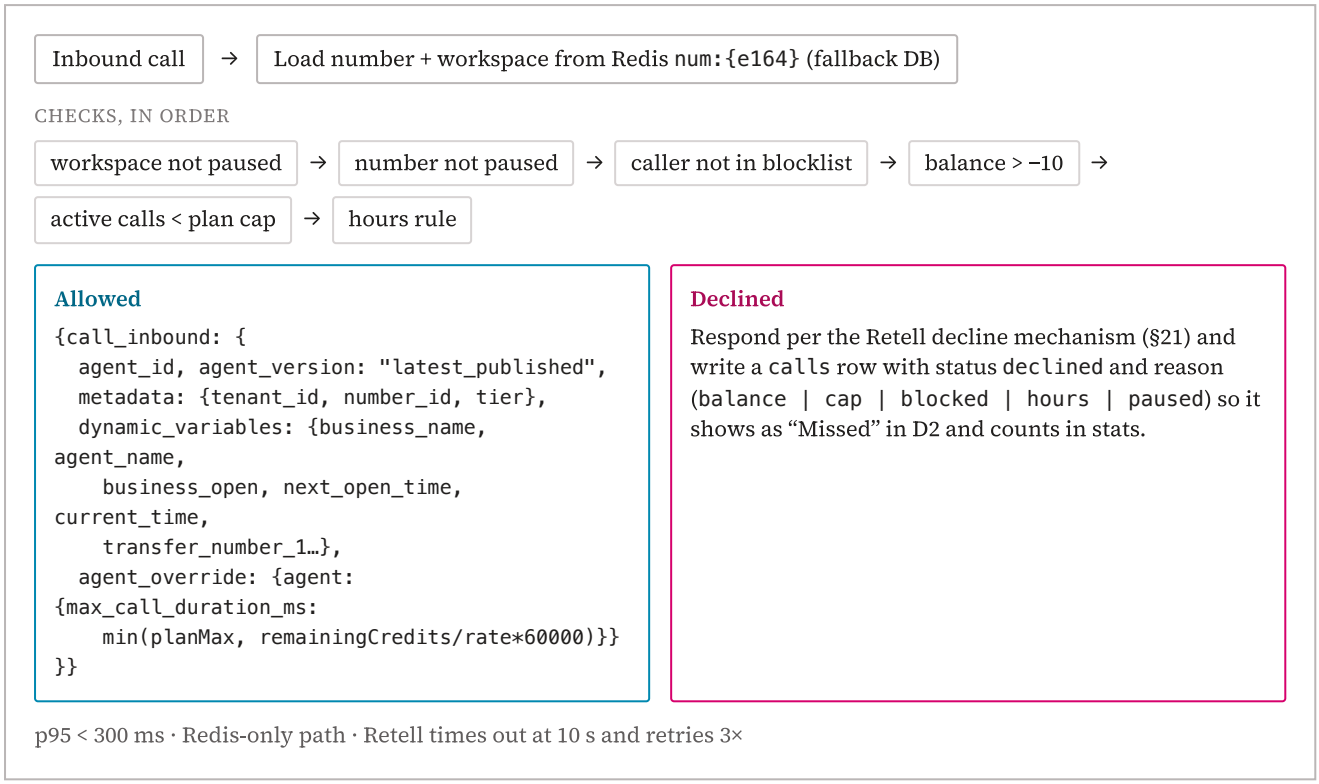


Figure 14 — Inbound call authorisation. Six ordered checks decide between the override response and a decline; both paths write a call row.

## 13.10 Flow — Call lifecycle webhooks

`POST /webhooks/retell` (agent-level webhook URL, all agents point here):

1. Verify signature → upsert `webhook_events` (unique `(retell, call_id, event)`) → if duplicate, 204.
2. `call_started` → upsert `calls` (status `in_progress`), `INCR active:{ws}` with TTL; SSE.
3. `call_ended` → duration, disconnection reason, transcript, recording URL → metering (§14.1) → `DECR active:{ws}` → balance recalculation → alert checks → SSE.
4. `call_analyzed` → `analysis_json`, `retell_cost_cents` → notifications (new message captured) → digest queue → SSE.
5. Any handler error → job retry (BullMQ, 5 attempts) — never 5xx to Retell after the event is stored.

## 13.11 Scheduled jobs

| JOB | SCHEDULE | LOGIC |
|---|---|---|
| `number-rental` | hourly | for numbers with `next_rental_at <= now`: debit 15 credits (or mark `rental_overdue` if balance ≤ −10), advance date, notify |
| `credit-expiry` | daily | expire lots past `expires_at`; ledger `expiry` rows; email 30/7 days before |
| `low-balance-alerts` | on ledger change + daily | thresholds 20% / 5% / 0 of last pack; dedupe per threshold per top-up cycle |
| `auto-topup` | on ledger change | if enabled and balance < threshold and no top-up in last 10 min → Stripe PaymentIntent off-session → on success credit lots; on failure email + banner, retry daily 3× |
| `reconcile-calls` | nightly | `POST /v2/list-calls` (last 48 h) vs `calls`; create/settle missing; compare Σ`combined_cost` vs expected COGS; alert > 5% drift |
| `voice-cache` | daily | `GET /list-voices` → `voices` table (tier mapping by provider) |
| `concurrency-monitor` | 5 min | record platform active calls; alert ops at 80% of Retell allowance |
| `number-release` | daily | numbers on workspaces at ≤ 0 balance for 30 days → release + email (warnings at 7/14/21) |
| `daily-digest` | per workspace TZ 07:00 | messages/appointments captured yesterday → email |
| `kb-sync` | on demand | poll Retell KB source status |

## 13.12 Environments and config

- `dev` (Retell test agents prefixed `dev-`), `staging`, `prod`. One Retell API key per environment; webhook URLs per environment. Stripe test/live keys. Feature flags via config table.
- Secrets in a secrets manager; SIP credentials encrypted with envelope encryption (KMS).

# 14 Credit metering and enforcement

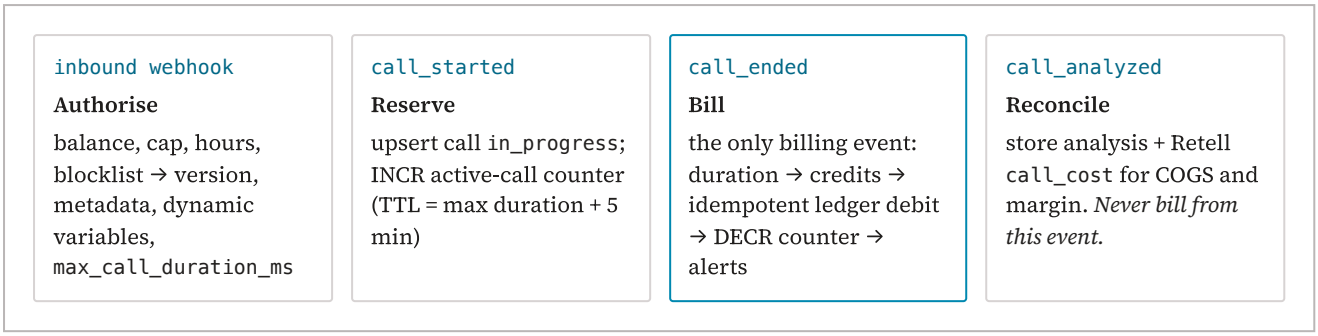


Figure 15 — The metering path. Four Retell events; only `call_ended` moves credits.

## 14.1 Events consumed

| RETELL EVENT | OUR ACTION |
|---|---|
| `inbound webhook` (per number, pre-call) | Authorise the call (§13.9): balance > −10 (grace), active calls < plan cap, hours/blocklist rules; respond with agent version, metadata, dynamic variables and `max_call_duration_ms` derived from remaining credits. If not authorised → decline (§21 for mechanism). 10-s Retell timeout, retried 3× — Redis-only path. |
| `call_started` | Create/upsert `calls` (status `in_progress`), increment tenant active-call counter (Redis, TTL = max duration + 5 min). Not fired if the dial fails. |
| `call_ended` | `duration_s = (end_timestamp − start_timestamp)/1000`; if `disconnection_reason` indicates not connected (dial failed / no answer / busy) → 0 charge. Else `credits = duration_s/60 × rate(tier)` rounded to 0.01. Ledger debit idempotent on `call_id`; decrement counter; update balance; alert checks. |
| `call_analyzed` | Store `call_analysis` and `call_cost` (Retell product breakdown + combined cost) → COGS tracking and margin dashboard. Never bill from this event. |
| `transfer_*` (Phase 2) | Split billing between agent leg and transferred leg. |

## 14.2 Rules

- **Idempotency:** unique `(call_id, event)`; replayed webhooks are no-ops.
- **Signature verification:** `X-Retell-Signature` verified with our API key on every request; reject otherwise.
- **Tenant isolation:** every webhook resolves tenant from `agent_id` / `to_number` mapping; a mismatch with `metadata.tenant_id` is logged and rejected.
- **Reconciliation (nightly):** `POST /v2/list-calls` for last 48 h vs `calls`; create/settle missing; compare Retell `combined_cost` totals vs expected COGS; alert on > 5% drift.
- **Outbound (Phase 2):** same balance/concurrency check in our API before `POST /v2/create-phone-call`; `max_call_duration_ms` set from remaining credits.

- **Zero balance UX:** number stays attached; inbound webhook declines; call logged as “Missed — paused (no credits)”; customer notified immediately; auto top-up resolves it if enabled.

### 14.3 Concurrency (platform-wide)

- Single Retell account = 20 free concurrent calls shared by all tenants. Per-plan caps (§5.2) enforced by us in the inbound webhook.
- Ops metric: platform peak concurrency per hour. Purchase Retell slots when 7-day peak > 80% of allowance; release when < 40% for 30 days. Slot cost is platform overhead.
- Cap-hit behaviour: decline + log “Missed (busy)” + optional voicemail agent (Phase 2).

## 15 Data model

| TABLE | KEY FIELDS |
|---|---|
| `users` | id, auth_provider_id, email, name, avatar_url, mfa_enabled, tos_accepted_at, created_at |
| `tenants` | id, name, country, timezone, address_json, website, plan_tier (trial/starter/growth/scale/pro/volume), concurrency_cap, kb_allowance, business_hours_json, stripe_customer_id, auto_topup_json {enabled, pack_id, threshold, payment_method_id}, onboarding_step, onboarding_completed_at, paused_at, recording_default, retention_days, created_at |
| `memberships` | id, tenant_id, user_id, role (owner/admin/editor/viewer), created_at; unique(tenant_id, user_id) |
| `invites` | id, tenant_id, email, role, token_hash, invited_by, expires_at, accepted_at |
| `audit_log` | id, tenant_id, actor_user_id, action, target_type, target_id, ip, meta_json, created_at |
| `notification_settings` | id, tenant_id, user_id (nullable = workspace-level), new_message (instant/daily/off), low_balance_email, low_balance_sms, weekly_summary |
| `templates` (config) | id, name, industry_tags[], brief_prefill_json, steps_json, analysis_fields_json, rules_json, default_tier, sample_audio_url, version |
| `agents` | id, tenant_id, name, template_id, industry, tier (standard/premium), primary_language, languages[], voice_id, voice_speed, voice_temperature, retell_agent_id, retell_llm_id, retell_kb_id, draft_version, published_version, status (draft/live/paused), max_call_duration_s, recording_enabled, addons_json, created_at |
| `agent_versions` | id, agent_id, version_no, source (generated/feedback/manual/regenerate), brief_json, prompt_text, greeting, closing, analysis_schema_json, brief_card_json, scenarios_json, changes_json, feedback_text, lint_json, retell_agent_version, created_by, published_at, created_at; unique(agent_id, version_no) |
| `voices` (cache) | retell_voice_id, provider, name, gender, accent, languages[], preview_url, tier, is_multilingual, updated_at |

| TABLE | KEY FIELDS |
|---|---|
| `knowledge_sources` | id, agent_id, type (url/file/text), name, retell_source_id, status (processing/ready/failed), size_bytes, error, created_at |
| `phone_plans` | id, tenant_id, agent_id, option (new/forward/sip), country, area_code, toll_free, existing_number, sip_json (encrypted), hours_mode, transfer_targets_json, allowed_countries[], status (pending/executed) |
| `phone_numbers` | id, tenant_id, agent_id, e164, nickname, type (retell_twilio/retell_telnyx/custom_sip), country, area_code, toll_free, hours_mode, hours_override_json, transfer_targets_json, allowed_countries[], blocklist[], inbound_token, rental_credits, next_rental_at, status (provisioning/active/paused/releasing/released), retell_meta_json, created_at |
| `calls` | id (retell call_id), tenant_id, agent_id, agent_version, number_id, channel (phone/web), direction, is_test, from_number, to_number, started_at, ended_at, duration_s, disconnection_reason, tier, rate, credits_charged, retell_cost_cents, transcript_text, transcript_json, recording_url, analysis_json, outcome, handled_at, note, status (registered/in_progress/ended/analyzed/declined), decline_reason |
| `credit_lots` | id, tenant_id, purchase_id, credits_total, credits_remaining, expires_at, created_at |
| `credit_ledger` | id, tenant_id, delta (numeric 12,2), balance_after, reason (trial/purchase/call/rental/kb/adjustment/expiry/refund/topup), ref_type, ref_id, lot_id, created_at |
| `purchases` | id, tenant_id, pack_id, amount_cents, currency, credits, stripe_session_id, stripe_payment_intent, invoice_url, status (pending/paid/abandoned/refunded), is_auto_topup, created_at |
| `packs` (config) | id, name, price_cents, currency, credits, plan_tier, concurrency_cap, kb_allowance, active |
| `webhook_events` | id, provider (retell/stripe/auth), event_type, external_id, payload_json, received_at, processed_at, error; unique(provider, external_id, event_type) |
| `jobs` | id, tenant_id, type, status, steps_json, error, created_at, updated_at |
| `alerts` | id, tenant_id, type, threshold, sent_at, channel, dismissed_at |

Balance = Σ `credit_ledger.delta`; cached in Redis `bal:{ws}`, rebuilt from the ledger on mismatch. Lots consumed FIFO by `expires_at` (call debits reference the lot).

# 16 API surface (portal ↔ backend)

All endpoints: `Authorization: Bearer <jwt>`, `X-Workspace-Id`. Errors: `{code, message, fields?[]}`. Pagination: `?cursor=&limit=`.

| AREA | ENDPOINTS |
|---|---|
| Workspaces | POST /workspaces · GET /workspace · PATCH /workspace · GET /workspace/checklist · GET /flags |
| Team | GET /members · POST /invites · POST /invites/:token/accept · PATCH /members/:id {role} · DELETE /members/:id · POST /workspace/transfer-ownership |
| Agents | POST /agents {template_id, industry} · GET /agents · GET /agents/:id · PATCH /agents/:id/draft (brief/voice/language/settings) · POST /agents/:id/generate · POST /agents/:id/feedback · POST /agents/:id/draft/select {version_no} · GET /agents/:id/versions · GET /agents/:id/versions/:v · PUT /agents/:id/versions/:v/prompt (advanced) · POST /agents/:id/publish · POST /agents/:id/pause · DELETE /agents/:id |
| Voices | GET /voices?lang=&gender=&tier=&multilingual= · POST /voices/:id/preview {text} |
| Knowledge | POST /agents/:id/knowledge · GET /agents/:id/knowledge · DELETE /agents/:id/knowledge/:sourceId |
| Test | POST /agents/:id/test/web {version_no} · POST /agents/:id/test/phone {to} (P2) |
| Numbers | PUT /agents/:id/phone-plan · GET /numbers · POST /numbers/buy · POST /numbers/import · GET /numbers/:id · PATCH /numbers/:id · DELETE /numbers/:id · GET /numbers/forwarding-guide?carrier= |
| Calls | GET /calls (filters) · GET /calls/:id · PATCH /calls/:id {handled, note} · GET /calls/:id/recording (signed URL) · POST /calls/export |
| Stats | GET /stats?range=today\|7d\|30d\|custom |
| Wallet | GET /wallet · GET /wallet/lots · GET /wallet/ledger · POST /wallet/checkout · PUT /wallet/auto-topup · GET /wallet/invoices · POST /wallet/portal-session |
| Notifications | GET/PUT /notifications/settings · GET /alerts/active · POST /alerts/:id/dismiss |
| Security | GET /sessions · DELETE /sessions/:id · GET /audit-log |
| Events | GET /events (SSE; topics: wallet, agent:{id}, call:{id}, activation:{id}) · GET /jobs/:id |
| Webhooks (no JWT) | POST /webhooks/retell · POST /webhooks/retell/inbound/:number_token · POST /webhooks/stripe · POST /webhooks/auth |

EXAMPLE — POST /AGENTS/:ID/TEST/WEB

```
// request
{ "version_no": 3 }

// response 201
{ "call_id": "Jabr9…", "access_token": "eyJ…", "rate": 1.0, "balance": 18.3 }

// errors: 402 NO_CREDITS, 403 EMAIL_UNVERIFIED, 429 TEST_RATE_LIMIT
```

EXAMPLE — POST /WALLET/CHECKOUT

```
{ "pack_id": "growth", "auto_topup": { "enabled": true, "threshold": 50 }, "agent_id": "ag_123"
}

// → { "checkout_url": "https://checkout.stripe.com/…", "purchase_id": "pur_…" }
```

# 17 Security, compliance and abuse controls

- **Auth & tenancy:** as §6; RLS in Postgres keyed by `workspace_id`; all Retell object ids mapped to tenants; web-call tokens minted server-side with an agent ownership check.
- **Recording & consent:** recording on by default in one-party jurisdictions; the greeting auto-includes a consent line when the workspace address is in a two-party state (CA, WA, FL, IL, MD, MA, MT, NH, PA, CT…; Quebec/Canada one-party — flag for legal review). Tenant can disable recording (Retell data-storage setting).
- **PII:** Retell PII-removal add-on toggle on Scale+ (medical/legal templates default on). Card numbers/SSN never collected (guardrail).
- **Outbound (Phase 2):** TCPA/CRTC — KYC (business verification), verified caller ID, DNC checks, calling windows, per-day caps. Not self-service on day one.
- **Fraud:** no phone number before first purchase; trial credits browser-only; allowed inbound country default = workspace country (+US/CA); test-call velocity limits; Stripe Radar; blocked disposable emails; manual review flag for > $1,000 first purchase.
- **Data retention:** transcripts/recordings 90 days default (30/90/365 configurable); Retell retention aligned; deletion propagates (`DELETE /delete-call`).
- **Secrets:** Retell, Stripe, Anthropic keys in a secrets manager; SIP creds envelope-encrypted; webhook signature verification mandatory; least-privilege service accounts.
- **HIPAA:** not offered in MVP (Retell requires Enterprise + BAA). The medical template ships with a “not for PHI” notice.
- **Backups:** Postgres PITR 7 days; the ledger is append-only (no UPDATE/DELETE grants for the app role).

# 18 Non-functional requirements

- Inbound webhook p95 < 300 ms (Redis path), hard cap 2 s; Retell times out at 10 s.
- Webhook processing exactly-once via `webhook_events` uniqueness; handler errors never surface as 5xx after storage.
- Balance drift vs Retell cost < 5% monthly (reconciliation alerts above that).
- Generation p95 ≤ 10 s; feedback patch p95 ≤ 8 s; wizard step navigation < 1 s; voice previews start < 1 s.
- Portal uptime 99.5%; agent availability bound by the Retell SLA (status surfaced on D7 and on our status page).
- Observability: OpenTelemetry traces (API + jobs), per-tenant call success rate, decline reasons, Retell error rates, LLM latency/cost, Stripe webhook lag; alerting to Slack.
- Accessibility WCAG 2.1 AA; i18n en/fr-CA; mobile web supported for all dashboard screens and the test call.

# 19 Phased delivery

### P0 — Spike

1 Week

SCOPE — Retell account; one agent via API; web call in browser; inbound number; all 3 webhooks + inbound webhook round-trip; confirm `call_cost` payload, decline mechanism and `max_call_duration_ms` override; benchmark GPT-5 mini vs GPT-4.1 mini on the receptionist template; Stripe Checkout + webhook in test mode

EXIT — Cost per test call matches the pricing page ±5%; decline path confirmed; end-to-end “brief → live number” done by hand

### P1 — MVP

1 Week

SCOPE — §6 auth (A1–A8), §8 wizard (W1–W7b) inbound only, §9 templates + generation + feedback + versions + lint, §10 D1–D7 (D3 Test tab without “call my phone”), §11 A/B/C numbers, §13 flows, §14 metering, alerts, auto top-up, daily digest, fr-CA

EXIT — 10 design-partner tenants live; median sign-up→live < 15 min; billing reconciles; zero cross-tenant access in pen test

### P2

1 Week

SCOPE — KB usage stats, SIP “re-verify”, transfer billing split, business-hours behaviour per number, “call my phone” test, web widget channel, forwarding test, outbound click-to-call + compliance gating, calendar/CRM tool calls (Google Calendar, HubSpot), multi-agent per number by hours, D8 admin console hardening

EXIT — Growth/Scale packs sold; transfer + KB billed correctly

**P3**

| | |
|---|---|
| SCOPE | Batch outbound campaigns, SMS follow-ups, number porting (ops process), agency/white-label tier, public API + API keys, additional countries (UK, AU, IN), chat agent channel |
| EXIT | — |

**Team assumption:** 2 full-stack + 1 part-time designer for P1; Pramod owns the Retell gateway and billing core; legal/compliance execution delegated (consent scripts, TCPA/CRTC review, Stripe MoR).

# 20 Decisions needed

1. Working name and domain.
2. Credit unit model (§5.5.1) and pack price points (§5.2). USD vs CAD; merchant of record entity.
3. Standard-tier LLM (GPT-5 mini vs GPT-4.1 mini) after the P0 benchmark.
4. Launch countries for numbers: US + CA only, or include India (Retell lists India Twilio/Telnyx; telephony rate to verify)?
5. Trial size (20 credits proposed) and whether to require a card at sign-up.
6. Credit expiry (12 months proposed) — check prepaid-credit consumer rules in target provinces/states.
7. Auth provider (Clerk recommended) and backend framework (NestJS vs FastAPI) — align with the existing Metacognixion platform stack.
8. Whether Starter allows exactly one agent and one number (recommended) and whether multiple agents are a Growth+ feature.
9. Whether the Viewer role may download recordings (recommended: no).
10. Support tooling (Intercom/Crisp) and status page vendor.
11. Portfolio positioning: under Metacognixion (NA brand) with Gritbyte delivering; cross-sell path into Sceniq/Solunary.

# 21 Items to verify against Retell during P0

- Exact inbound-webhook response to **decline** a call vs. override to a fallback agent, and whether a declined call generates any cost.
- Whether `agent_override.agent.max_call_duration_ms` is honoured from the inbound webhook (used for overdraft protection).
- `call_cost` object shape and units in `call_analyzed` / `GET /v2/get-call`.
- Toll-free and non-US telephony per-minute rates (the pricing page shows a per-country selector).

- Retell “workspaces”/sub-accounts — if available, evaluate one workspace per tenant for isolation (affects concurrency and KB free-tier accounting).
- `language: multi` behaviour with each voice provider; which voices are truly multilingual; exact locale list for the Main-language select.
- Agent version tags (`latest_published`, custom tags) for number binding, and the publish API call.
- Knowledge base ingestion limits (file size, URL crawl depth) and refresh behaviour.
- TTS preview endpoint for “Hear my greeting” (else use provider preview clips only).
- Website widget availability and whether it counts against concurrency.
- Whether `post_call_analysis_data` supports enum/number types as assumed in §9.8.

# A Retell endpoints used

| PURPOSE | ENDPOINT |
|---|---|
| Voices | `GET /list-voices` |
| Agent lifecycle | `POST /create-agent`, `PATCH /update-agent`, `GET /get-agent`, publish/version APIs |
| LLM config (prompt, model, tools) | `POST /create-retell-llm`, `PATCH /update-retell-llm` |
| Knowledge base | `POST /create-knowledge-base`, add/delete sources, get status |
| Numbers | `POST /create-phone-number`, `POST /import-phone-number`, `PATCH /update-phone-number`, `DELETE /delete-phone-number`, `GET /list-phone-numbers` |
| Calls | `POST /v2/create-web-call`, `POST /v2/create-phone-call`, `GET /v2/get-call`, `POST /v2/list-calls`, `DELETE /delete-call` |
| Events in | Agent webhook (`call_started`, `call_ended`, `call_analyzed`, `transfer_*`), number `inbound_webhook_url` |

# B Prompt template skeleton (generated per agent)

```
## Identity
You are {agent_name}, the {role} for {business_name}, {business_description}. Personality:
{persona}.

## Goal
{primary_goal}. Success = {success_definition}.

## Business facts
Hours: {hours} · Address: {address} · Services: {services} · Pricing basics: {pricing}
FAQ: {faq_pairs}

## Conversation guide
1. Greet in {primary_language}; switch languages if the caller does ({languages}).
2. Identify the caller's need in one question.
3. {template_specific_steps}
4. Confirm captured details back to the caller before ending.

## Rules
- Never: {never_rules}
- If unsure, say you'll have someone call back; never invent facts.
- Keep answers under 2 sentences unless asked for detail.

## Hand-off
{handoff_rules} → use transfer to {transfer_targets}; otherwise take a message.

## Ending
{closing_line}

## Platform rules (fixed)
{platform_guardrails — §9.4}
```

Post-call analysis fields generated alongside: `{template_fields}` + `caller_intent`, `follow_up_required`.

# C Worked cost examples

| SCENARIO | RETELL COGS / MO | CREDITS / MO | PACK | REVENUE / MO | GROSS MARGIN |
|---|---|---|---|---|---|
| Clinic, 300 min inbound, Standard, 1 number | 300×$0.11 + $2 = $35 | 300 + 15 rental = 315 | Growth ($149/500cr) ≈ 0.63 pack | ≈ $94 | ≈ 63% |
| Same, Premium voice | 300×$0.16 + $2 = $50 | 450 + 15 = 465 | ≈ 0.93 Growth pack | ≈ $139 | ≈ 64% |
| Home-services, 1,200 min, Standard, 2 numbers, KB | 1,200×$0.11 + $4 (+$8 KB if >10 platform-wide) = $136–144 | 1,200 + 30 = 1,230 | ≈ 0.82 Scale pack | ≈ $327 | ≈ 56–58% |

# D Copy deck (key strings, en; fr-CA to follow)

| KEY | TEXT |
|---|---|
| `wizard.w1.title` | What should your AI agent do? |
| `wizard.w2.title` | Tell us about your business |
| `wizard.w2.hint` | Write like you're briefing a new receptionist. Short answers are fine. |
| `wizard.w2.generate` | ✦ Generate my agent |
| `wizard.w3.title` | Choose a voice and languages |
| `wizard.w5.title` | Test your agent |
| `wizard.w5.improve.placeholder` | e.g. "Ask for the postal code before confirming" |
| `wizard.w6.title` | How should callers reach your agent? |
| `wizard.w7.cta` | Pay {amount} & go live |
| `wallet.paused.banner` | Your agent is paused — you're out of credits. Top up to resume answering calls. |
| `calls.empty` | No calls yet. Call {number} to hear your agent. |
| `publish.confirm` | Publishing updates your live agent immediately. Continue? |

# E Sources

Checked 10 Sep 2026

- Retell pricing page (component rates, monthly items, FAQ on billing granularity): retellai.com/pricing
- Retell webhook overview (events, payloads): docs.retellai.com/features/webhook-overview
- Retell inbound call webhook (override / reject / dynamic variables): docs.retellai.com/features/inbound-call-webhook
- Retell create web call (access token, agent_override): docs.retellai.com/api-references/create-web-call
- Retell create / import phone number: create-phone-number, import-phone-number
- Retell Twilio elastic SIP trunking guide: docs.retellai.com/deploy/twilio
- Third-party market pricing references (Cloudtalk, Layer3 Labs, Rezora, Ainora): $0.11–0.18/min realistic Retell all-in; SMB resale $0.25–0.50/min.